

UNIVERSITY OF SOUTHAMPTON

THESIS

Analysis of CubeSat drag using Direct Simulation Monte Carlo (DSMC) Method

Author (Corresponding):
Debjyoti Sengupta

Supervisor(s):
Dr. Minkwan Kim, Dr. Alexander Wittig

Word Count: 10450 Email ID: ds5n19@soton.ac.uk

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Faculty of Engineering and Physical Sciences, University of Southampton, Southampton,
United Kingdom – SO17 1BJ

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“You have to dream before your dreams can come true.”

In memory of my late Grandpa.

“We choose to go to the Moon in this decade and do the other things, not because they are easy, but because they are hard.....”
- John F. Kennedy

I dedicate this thesis to the global space community, my family & friends.

Declaration of Authorship

I, Debjyoti SENGUPTA, declare that this thesis and the work presented in it are my own and has been generated by me as the result of my own original research.

I confirm that:

1. This work was done wholly whilst in candidature for a degree at this University;
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3. Where I have consulted the published work of others, this is always clearly attributed;
4. Where I have quoted from the work of others, the source is always given, With the exception of such quotations, this thesis is entirely my own work;
5. I have acknowledged all main sources of help;
6. Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself;
7. None of this work has been published before submission.

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Debjyoti Sengupta

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List of Abbreviations

AIAA – American Institute of Aeronautics and Astronautics

ASCII – American Standard Code for Information Interchange

CFD – Computational Fluid Dynamics

DSMC – Direct Simulation Monte Carlo

EM – Electromagnetic

ESA – European Space Agency

EUV – Extreme Ultraviolet

K – Kelvin, unit for temperature

MB – Megabyte

MSISE – Mass Spectrometer and Incoherent Scatter Extended

NASA – National Aeronautics and Space Administration

SMAD – Space Mission Analysis and Design

U – Unit

UV - Ultraviolet

VHS – Variable Hard Sphere

VSS – Variable Soft Sphere

kg – kilogram, unit for mass

m – metre, unit for length

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Abstract

DSMC is a statistical and particle-based method in which several representative particles are traced in space and time. It is a numerical solution method to solve the dynamic equations for rarefied gas flow by at least thousands of simulated molecules. More specifically, this method is reliant on the numerical solution obtained from the Boltzmann kinetic equation for gases. DSMC is believed to be the most useful for the analysis of kinetic nonequilibrium gas flows. Nonequilibrium aerothermodynamics is an integral component in the implementation of atmospheric re-entry. The ability to simulate non-equilibrium flows using particle simulation methods has progressed rapidly during the last several decades and has fundamentally changed the aerospace design process. This project will use the DSMC method to simulate the atmospheric re-entry of a CubeSat where the breakdown of the continuum assumption is occurring and analyse the aerodynamic drag acting on the spacecraft by comparing different values of the drag coefficient for different parameters like angle of attack, geometry of the CubeSat, solar activity and thermochemical nonequilibrium.

Keywords: DSMC, Drag Coefficient, Temperature, Density, Rarefied Gas Dynamics, Nonequilibrium, Re-entry, Hypersonic, Aerothermodynamics, CubeSat.

1. Introduction

1.1. Motivation for Project

The use of CubeSat-like small satellites is growing exponentially nowadays, pushing towards missions of increased capability, including Earth imaging, commercial communications, and astronomical observations. As such, they might require payloads that could survive the re-entry conditions and reach the ground with the intention of retrieval, posing risks to population and properties [1-3]. The possibility of demise and ground impact poses many challenges from the modelling standpoint because of the uncertainties associated with both the aerodynamic and the aerothermodynamic models of the spacecraft. Using the DSMC method, the project will accurately predict the aerodynamic drag of a CubeSat in the rarefied atmosphere at a re-entry altitude ranging between 80 to 100 km at Mach 25 hypersonic flow.

1.2. Literature Review

1.2.1. CubeSat – A Small Satellite

CubeSats have become the next great innovation in the space sector, and these platforms are being used for a variety of space-related applications like education, Earth observation, remote sensing, science and defence. The term CubeSat is used to describe a small cube-shaped satellite whose basic unit form is a 10 cm edge cube, namely 1U or 1 “unit”[4]. These units can be combined to form larger CubeSat configurations like 2U, 3U, 6U and so forth. The largest CubeSat configuration on records to date has been 27U [5]. CubeSats are required to abide by the standards laid down in the CubeSat Design Specification [6]. These standards also include flight safety guidelines that ought to be complied with while designing the satellite. By the end of 2018, more than one thousand CubeSats developed by various organisations based in 59 countries have been launched [6, 7], as illustrated by Figure 1.

The motivation for the initial development of CubeSats was to provide a practical and hands-on experience to students in space engineering and allow them to work on the entire cycle of a space project. When the first CubeSats were launched in the first decade of 21st century, they were meant to meet amateur demands. However, it was soon realised that these small satellites have great potential and can be used for different applications, ranging from testing of technologies [8], astronomical observations [9-11], and space weather [12].

CubeSats serve as great platforms to test and implement miniaturised versions of technologies like communication antennas [13], propulsion units [14], and electrical power subsystems [15]. These technological solutions are contributing to the advent of the next generation in spaceflight. Advanced engineering solutions tested using CubeSats have allowed for the evolution in the design of passive thermal control

systems, thus allowing CubeSats to venture into harsher space environment found in deep space and beyond and improving the scope of sample return missions [16].

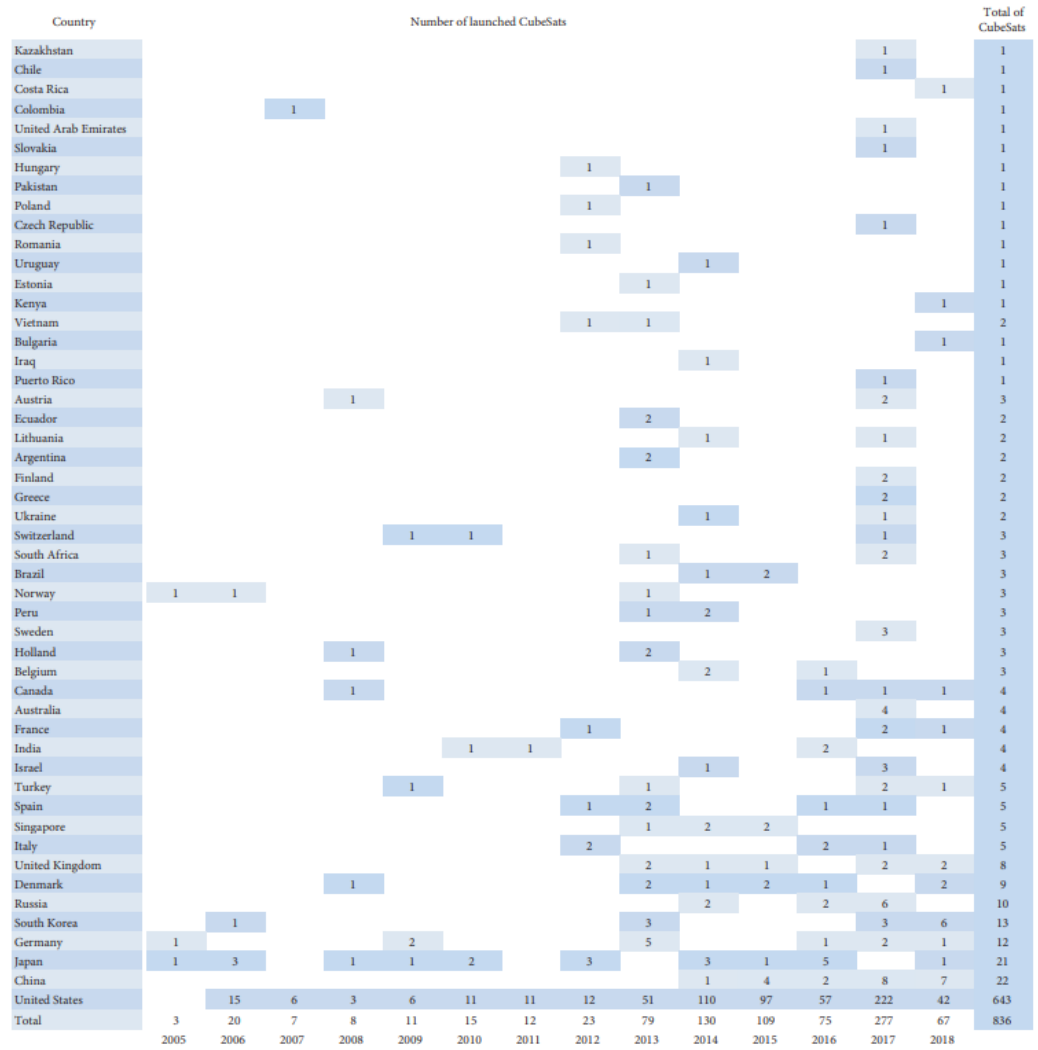


Figure 1: Number of CubeSats launched per year per country from 2005 to May 31, 2018 [6].

1.2.2. Rarefied Gas Dynamics, Nonequilibrium and Atmospheric Re-Entry

1.2.2.1. Rarefied Gas Dynamics

A gas flow may be modelled at either the macroscopic or the microscopic level. The macroscopic model regards the gas as a continuous medium and the description is in terms of variations in flow properties like the velocity, density, pressure and temperature dependent on space and time. The Navier-Stokes equations provide the conventional mathematical model of a gas as a continuum. The macroscopic properties are the dependent variables in these equations while the independent variables are the spatial coordinates (x, y, z) and time t .

The microscopic model recognises the particulate structure of the gas as an uncountable set of discrete molecules and ideally provides information on the position, velocity and state of every molecule always. The expression for the Boltzmann equation for a simple dilute gas can be written as [17]:

$$\frac{\partial}{\partial t}(nf) + \mathbf{c} \cdot \frac{\partial}{\partial \mathbf{r}}(nf) + \mathbf{F} \cdot \frac{\partial}{\partial \mathbf{c}}(nf) = \int_{-\infty}^{\infty} \int_0^{4\pi} n^2(f^* f_1^* - f f_1) c_r \sigma d\Omega d\mathbf{c}_1 \quad 1$$

The Boltzmann equation is the representation of the microscopic mathematical model for rarefied gases. In Eq. 1, n is the number density, f is the probability density function which is a function of phase space, velocity and time, $\mathbf{c}$ is the velocity distribution function, $\mathbf{r}$ is the spatial element, and $\mathbf{F}$ is an external force acting on each particle. The difference between f and f^* should be noted as the * stands for post-collision state. The Earth's atmosphere should be considered as a gas mixture. The model can be extended to gas mixtures using mixing rules, like Gupta's mixing rule [18]. The Boltzmann equation then becomes a set of simultaneous equations and particular species may be represented by the subscripts p or q . The Boltzmann equation for species p of the mixture can therefore be written:

$$\begin{aligned} \frac{\partial}{\partial t}(n_p f_p) + \mathbf{c}_p \cdot \frac{\partial}{\partial \mathbf{r}}(n_p f_p) + \mathbf{F} \cdot \frac{\partial}{\partial \mathbf{c}}(n_p f_p) \\ = \sum_{q=1}^s \int_{-\infty}^{\infty} \int_0^{4\pi} n_p n_q (f_p^* f_{1q}^* - f_p f_{1q}) c_{rpq} \sigma_{pq} d\Omega d\mathbf{c}_{1q} \end{aligned} \quad 2$$

The term on the right-hand side of the Boltzmann equation expressions Eqs. 1 and 2 is called the collision term. Its integral form contrasts with the partial differential form of the terms expressing space and time dependence of nf , and is responsible for much of the mathematical difficulty associated with the Boltzmann equation. However, nf is the only dependent variable in the equation, thus giving the Boltzmann equation an advantage when compared to the Navier-Stokes (NS) equations [17, 19].

One of the most important developments in the history of statistical mechanics was the derivation by Ludwig Boltzmann of an integrodifferential equation that describes the approach to equilibrium of the one-particle phase-space density function $F(\mathbf{r}, \mathbf{p})$ of a rarefied gas. Some of the fundamental results obtained from Boltzmann's equation are:

1. The time-independent solution of the equation is the Maxwell-Boltzmann distribution $F(\mathbf{r}, \mathbf{p}) = C \exp \left\{ -\frac{\left[\frac{\mathbf{p}^2}{2m} + U(\mathbf{r}) \right]}{kT} \right\}$.
2. It was possible to define an entropy function with the characteristics that:
 - a. The entropy function is equal to the logarithm of the phase-space volume available to a system with a given phase-space density[17].
 - b. The entropy increases steadily until the phase-space density function becomes equal to the Maxwell-Boltzmann distribution.
 - c. At equilibrium, the Boltzmann entropy function is equal to the known thermodynamic entropy of an ideal gas[20].

The macroscopic properties may be identified with average values of the appropriate molecular quantities at any location in a flow. They may be defined if there are enough molecules within the smallest significant volume of a flow. This condition is satisfied by normal gas flow and results from the molecular model can be expressed in terms of continuum flow properties. A common point of debate and question is the necessity of a separate microscopic molecular model when the continuum assumption of the Navier-Stokes equation is helpful. The answer lies with mean free path λ – the average distance travelled by the molecules between collisions. The degree of rarefaction of a gas is generally expressed through the Knudsen number Kn , which is the ratio of the mean free path to the characteristic dimension L , i.e.

$$Kn = \frac{\lambda}{L} \quad 3$$

where L can be defined as the scale length of the macroscopic gradients and can be expressed for representation as:

$$L = \frac{\rho}{d\rho/dx} \quad 4$$

where ρ is the density of the gas and x is the Cartesian space element in x -axis. The standard requirement for Navier-Stokes equations to hold true is that the Knudsen number should be less than 0.1. When Knudsen number has the value of zero, the Navier-Stokes equations reduce to inviscid Euler equations. On the other hand, at infinite value for Knudsen number, we enter the region of free-molecule flow regime, described by the mathematical model of collisionless Boltzmann equation[19].

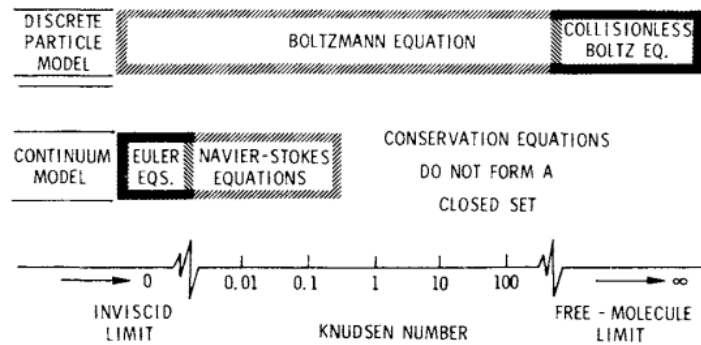


Figure 2: The Knudsen number limits on the mathematic models used to describe neutral gas flows [21].

At low densities, the Knudsen number becomes larger. As a result, in a rarefied gas flow with shock waves and boundary layers, the proportion of the flow that is viscous in nature increases. As the density falls, the rate of collision between molecules becomes too low, causing the continuum model to break down. It is because of this reason, engineers and physicists do not use computational fluid dynamics (CFD) calculations and resort to the DSMC method, as far as rarefied gas flows are concerned.

Bird has studied on the occurrence of this breakdown in gaseous expansions and stated the “breakdown parameter” [19, 22, 23]. The breakdown parameter for a steady rarefied gas flow can be expressed as:

$$P = \frac{\pi^{1/2}}{2} s \frac{\lambda}{\rho} \left| \frac{d\rho}{dx} \right| = \frac{\pi^{1/2}}{2} s \cdot Kn \quad 5$$

where s is the speed ratio. The breakdown parameter defines a non-dimensionless value, when satisfied, signifies the breakdown in the continuum assumption. The initial symptom of the breakdown is given by anisotropic pressure tensor. The initial breakdown in steady expansions have been found to correlate with a value of P of approximately 0.02, which corresponds to a Knudsen number between 0.1 to 1 [19].

1.2.2.2. Nonequilibrium

The zeroth law of thermodynamics can be quoted, according to Buchdahl, as follows:

“.....if each of two systems is in equilibrium with a third system then they are in thermal equilibrium with each other. [24]”

According to the above definition, we can state that when a gas is at thermal equilibrium, individual molecules gain and lose energy through collisions but the resultant change in energy is equal to zero. We can observe thermal equilibrium when all active modes of energy (translational, rotational and vibrational) are the same, which is in accordance with the principle of equipartition of energy [25, 26].

Similarly, chemical equilibrium can be defined, according to Atkins et. al:

“...it is a state in which both the reactants and products are present in concentrations which have no further tendency to change with time, so that there is no observable change in the properties of the system [27].”

Referring to the question on why we should consider a gas at the molecular level in Section 1.2.2.1. Rarefied Gas Dynamics, the molecular approach certainly provides us with a deeper insight into different types of gas flows. However, under boundary-layer and shock wave conditions, there is either not enough time or physical spatial volume for enough intermolecular collisions to occur to maintain the gas in the equilibrium state[25]. These conditions are referred to as nonequilibrium gas flow.

Beginning with the Boltzmann equation, moments corresponding to the molecular mass, momentum and energy can be taken to obtain a set of molecular conservation equations. These equations should include the general averages over the local velocity distribution function and are accurate for any degree of nonequilibrium. In the limit of near-equilibrium, through Chapman-Enskog analysis[23], the precise velocity distribution function is obtained that reduces the conservation equations to

the Navier-Stokes equations. The distribution function is a first-order perturbation to a Maxwell-Boltzmann equilibrium distribution, where the deviation is defined by the Knudsen number. Since the Chapman-Enskog distribution functions is an accurate representation of molecular model of gas, the Navier-Stokes equations in near-equilibrium (very small Knudsen number) limit are accurate as well[17, 19].

In hypersonic rarefied gas dynamics, the gas is not given the luxury of the required time to achieve the state of equilibrium. A perfect example is the flow across a shock wave, where the temperature and pressure are rapidly increased within the shock front. Due to the change in pressure and temperature, the vibrational and chemical equilibrium properties undergo change. To achieve the new equilibrium properties, a set number of intermolecular collisions are required. By the time equilibrium is achieved, the shock front moves by a certain distance downstream, causing the formation of a nonequilibrium region.

Vibrational nonequilibrium can be defined as the result of rate of change of vibrational energy of gas caused by molecular collisions at a given temperature. The above definition can be expressed in terms of the expression below, called the vibrational rate equation:

$$\frac{dE_{vibrational}}{dt} = \frac{1}{\tau} (E_{vibrational}^{eq} - E_{vibrational}) \quad 6$$

where $E_{vibrational}$ is the local instantaneous nonequilibrium value of vibrational energy mode, $E_{vibrational}^{eq}$ is the equilibrium value, and τ is the vibrational relaxation time, which is a function of local pressure and temperature[25, 28].

Similarly, when temperature of the gas composed of pre-defined species is instantaneously increased, change in composition is demanded to achieve the new equilibrium state. The rate of change in the molecular species composition in the gas due to molecular collisions and chemical reactions is defined as chemical nonequilibrium. The chemical nonequilibrium can be quantified by expressing them a set of first-order ordinary differential equations (ODEs), called the chemical rate equations. They have been expressed below:

$$\frac{d[X_i]}{dt} = (\chi_p - \chi_r)k_f \prod_i [X_i]^{\chi_r} \dots \dots \dots Forward \text{ rate} \quad 7$$

$$\frac{d[X_i]}{dt} = -(\chi_p - \chi_r)k_b \prod_i [X_i]^{\chi_p} \dots \dots \dots Reverse \text{ rate} \quad 8$$

$$\frac{d[X_i]}{dt} = (\chi_r - \chi_p) \left\{ k_f \prod_i [X_i]^{\chi_r} - k_b \prod_i [X_i]^{\chi_p} \right\} \dots \dots \dots Net \text{ Rate} \quad 9$$

where X_i is the molecular species, χ_r and χ_p are the stoichiometric mole numbers of the reactants and products respectively, and k_f & k_b are the rate constants for

forward and reverse reactions. The rate constant k depends on the activation energy E_a and temperature T , and this relation can be given by the Arrhenius equation:

$$k = C e^{-E_a/RT} \quad 10$$

where C is the pre-exponential factor and R is the universal gas constant.

Chemical nonequilibrium occurs when the characteristic time for the chemical reactions to reach local equilibrium is of the same order as the characteristic time of the flow. Similarly, thermal nonequilibrium occurs when the characteristic time for various internal energy modes to reach local equilibrium is of the same order as the characteristic time of the flow. Thermo-chemical nonequilibrium effects prevail in hypersonic rarefied gas flows since thermo-chemical changes are the results of collisions between constituent particles [29].

1.2.2.3. Atmospheric Re-Entry

The total energy of the spacecraft reduces proportionally to the rate at which it does work on the atmosphere. The passage of the spacecraft heats the atmosphere by the essentially adiabatic compression of the bow shock wave and by friction in the boundary layer over the surface of the vehicle. To ensure that the spacecraft is intact, the thermal load generated during re-entry must be properly managed. Heat shields are used as thermal protection, where the shield faces the flow and is subject to convective and radiative heat transfer[30, 31]. The above can be illustrated by the dimensionless equation of motion

$$\frac{1}{g_E} \dot{V} = -\frac{D}{mg_E} - \sin \beta \quad 11$$

where g_E is Earth's gravity, V is entry velocity of the spacecraft, β is the angle of attack, m is the mass of spacecraft during re-entry, and D is the aerodynamic drag force. Multiplying Eq.11 by V yields the rate of energy change of the spacecraft

$$m \left(V \frac{dV}{dt} + g_E V \sin \beta \right) = m \left(\frac{1}{2} \frac{dV^2}{dt} + g_E \frac{dz}{dt} \right) = -\frac{1}{2} C_D \rho_\infty V^3 S \quad 12$$

where z is the altitude of the spacecraft.

1.2.3. Aerodynamic Drag in Rarefied Gas

Flight is the motion of an object in or through the atmosphere of the Earth or through space. The major constraints for spaceflight are the gravitational force and the aerodynamic drag. The aerodynamic drag which is caused by the interaction of an object in motion with the displaced air particles. The drag of an aerodynamic body is given by:

$$D = \frac{1}{2} C_D \rho_\infty U_\infty^2 S \quad 13$$

where C_D is the drag coefficient, ρ_∞ is the free stream density, U_∞ is the free stream velocity and S is the cross-sectional area of the spacecraft.

In the expression Eq.13, it can be observed that the nondimensional drag coefficient has a dependence on the geometry and on the angle of incidence [32]. In the case of satellite, we can state that from Eq.13, ρ_∞ is the free stream density of the atmosphere at the satellite's position and S is the cross-sectional area of the satellite normal to its direction of motion. The equation stays valid in the rarefied gas regime, where the Knudsen number Kn is much greater than 1.

Let us consider a surface element of a solid body past which flows a steady rarefied gas flow with a constant macroscopic velocity U_∞ . Using the coordinate system (x, y, z) in which the x -coordinate is directed along the internal normal to the element, it is possible to represent the Maxwellian distribution function of incident molecules in the form

$$F = n_\infty (2\pi RT_\infty)^{-\frac{3}{2}} \exp \left\{ -\frac{[(c_1 - U_\infty \sin \beta)^2 + (c_2 + U_\infty \cos \beta)^2 + c_3^2]}{2RT_\infty} \right\} \quad 14$$

as described by Shidlovskiy [26]. The symbols c_1 , c_2 and c_3 denote components of molecular velocities on the corresponding axes, and the subscript ∞ stands for incoming free stream molecular flow. The number of molecules incident on a unit area of the surface element per unit time is

$$N_i = \int_0^{+\infty} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} F c_1 dc_1 dc_2 dc_3 \quad 15$$

Substituting the expression for F in Eq. 15 and integrating, we get

$$N_i = n_\infty \sqrt{\frac{RT_\infty}{2\pi}} \{e^{-s^2} \sin^2 \beta + \sqrt{\pi s} \times \sin \beta [1 + \operatorname{erf}(s \sin \beta)]\} \quad 16$$

Here we have defined the molecular speed ratio s by

$$s = \frac{U_\infty}{\sqrt{2RT_\infty}} = \sqrt{\frac{\gamma}{2}} M_\infty \quad 17$$

where $\gamma = c_p/c_v$ is the ratio of specific heats of the air at constant pressure and constant volume, $M_\infty = U_\infty/a_\infty$ is the Mach number of the flow and the function $\operatorname{erf} x$ is the probability integral, or the error function [33]:

$$\operatorname{erf} x = \frac{2}{\sqrt{\pi}} \int_0^x e^{-x^2} dx \quad 18$$

The impact of molecules incident on the surface exerts a force in the form of a normal pressure and tangential stress, p and τ respectively. These terms can be determined by calculation of the corresponding components of momentum. Since the macroscopic motion in the vicinity of the element of area is symmetrical with respect to the xy -plane, the tangential stress component directed along the z -axis

should be zero, and we can state the same for the stress directed along y-axis. The effective resultant pressure due to reflected molecules p_w can be expressed as:

$$p_w = \frac{m}{2} \sqrt{2\pi RT_w} N_i \quad 19$$

We obtain Eq. 19 assuming that the macroscopic velocity is zero for the Maxwellian distribution at wall temperature T_w , thus setting the value of s in Eq. 17 to zero. Because of the steady nature of the process, the number of reflected molecules N_r is equal to the number of incident molecules, N_i .

The total values of the pressure and tangential stress are found to be

$$p = p_i + p_r \quad 20$$

From Eqs. 16 – 20 we finally get

$$p = \rho_\infty RT_\infty \left\{ \left(\frac{2-\theta}{\sqrt{\pi}} s \sin \beta + \frac{\theta}{2} \sqrt{\frac{T_w}{T_\infty}} \right) e^{-s^2 \sin^2 \beta} + \left[(2-\theta) \left(s^2 \sin^2 \beta + \frac{1}{2} \right) + \frac{\theta}{2} \sqrt{\frac{\pi T_w}{T_\infty}} s \sin \beta \right] \times [1 + \operatorname{erf}(s \sin \beta)] \right\} \quad 21$$

where θ is the diffuse reflection coefficient which defines the average fraction of all the molecules that are reflected in a random manner [19, 20]. Using Eqs. 21, which describe the free-molecular forces on an elementary area of a solid surface, it is possible to find the drag of simple shapes in such a flow.

In the case of a flat plate [34], the projection of the total force on the x-axis yields the drag force. The drag force can be given by:

$$D = S \{ [p(\beta) - p(-\beta)] \sin \beta + [\tau(\beta) + \tau(-\beta)] \cos \beta \} \quad 22$$

The drag coefficient C_D in Eq. 13 can be expressed as:

$$C_D = \frac{2D}{\rho_\infty U_\infty^2 S} = \frac{2}{\sqrt{\pi} s} [2(1-\theta) \sin^2 \beta + \theta] e^{-s^2 \sin^2 \beta} + 2 \sin \beta \left\{ \left[2(1-\theta) \sin^2 \beta + \theta \right] + \frac{2-\theta}{2s^2} \right\} \operatorname{erf}(s \sin \beta) + \frac{\theta \sqrt{\pi}}{s_w} \sin \beta \cos \beta \quad 23$$

Eqs. 22 – 23 are like the analytical solution obtained by Shidlovskiy for drag acting on a flat plate at a given angle of attack β . From Eq. 23, the coefficient of drag for a flat plate of surface area 1 metre² in perpendicular flow is found to be 1.98. This result is consistent with the value obtained by Hoerner in 1965[35]. This would mean that the drag coefficient for the surface of 1U CubeSat will be 0.198.

1.2.4. Earth's Atmosphere

It is commonly understood within the scientific community that the Earth's atmosphere is comprised by some concentric spherical layers. These layers are notably characterised by their different temperature gradients. The layers are separated from each other by quite thin transition zones and these zones are the region where the temperature gradient changes.

As seen in Figure 3, the atmosphere has four layers – troposphere, stratosphere, mesosphere and thermosphere. The fifth layer, known as the exosphere, starts beyond the top limit of the thermosphere and is off the scale of Figure 3. The regions of most interest for those related to space science and engineering are the mesosphere and the thermosphere. The mesosphere starts just above the stratopause, the transition zone between the mesosphere and stratosphere. This layer extends to an altitude of approximately 90 km and the temperature gradient is negative. Atmospheric re-entry from space for vehicles is believed to begin at the altitudes ranging from 100 km to 80 km. The thermosphere starts just above the mesopause and extends to the altitude of 600 km. The temperature level in these altitudes depends upon the degree of solar activity. The thermosphere is also believed to be the region of low earth orbit (LEO). The temperature distribution, according to the 1976 U.S. Standard Atmosphere model [36], for the first 100 km can be represented by Figure 4.

Atmospheric composition i.e., the distribution of the species present in the atmospheric air depends on altitude. The composition of the atmosphere around the concerned altitudes is comprised of 78% nitrogen, 21% oxygen and 1% other gasses. Diffusive separation of species dominates in the given altitudes. It is observed that there is a decrease in the presence of molecular species (N_2 and O_2) as you go through the mesosphere and thermosphere. However, solar activity causes dissociation reactions leading to formation of species like atomic oxygen, atomic nitrogen and nitrous oxide. These species can result into harmful effects on the surface of the spacecraft [37].

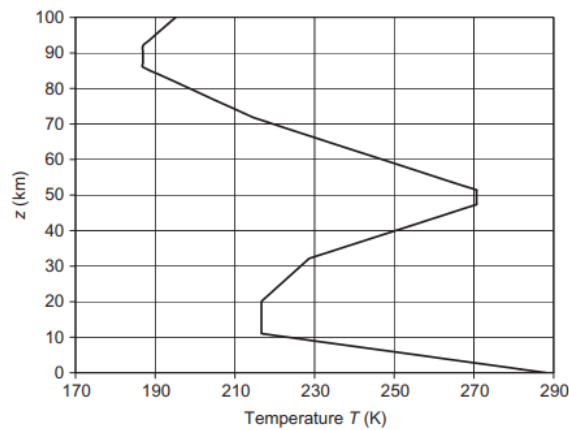


Figure 3: Variation in temperature at first 100 km of Earth's atmosphere [37].

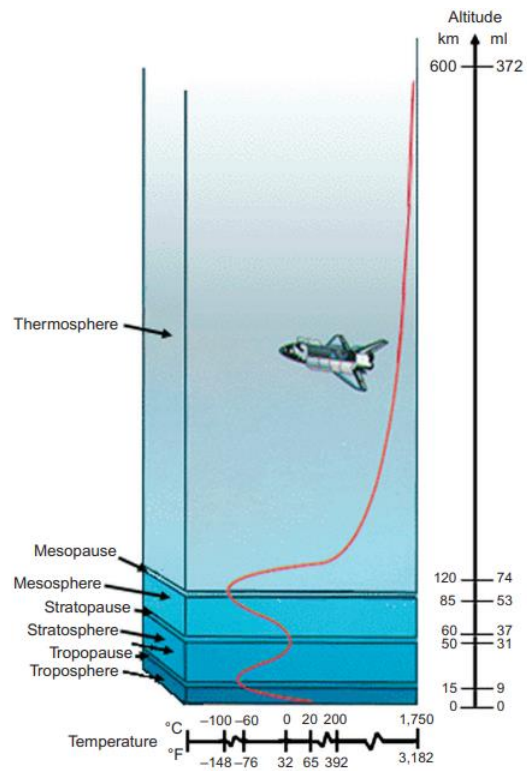


Figure 4: Relation between temperature and altitude of the Earth's atmosphere [37].

2. Methodology

2.1. Direct Simulation Monte Carlo (DSMC) Method

The Direct Simulation Monte Carlo (DSMC) method was conceived by Graeme Bird, a professor of aeronautical engineering at the University of Sydney, Australia. The first publication in 1963 demonstrated the DSMC method as an alternative to molecular dynamics (MD) for a hard-sphere (HS) gas [38]. Since then, an entire field of research has emerged with innumerable publications on the method and its applications to scientific and engineering problems. The DSMC method has become a powerful and widespread technique for the simulation of nonequilibrium flow where the molecular nature of the gas needs to be taken into consideration [17].

The method is similar to MD method but the essential difference is that the intermolecular collisions are dealt with a probabilistic approach rather than a deterministic approach [39]. This computational technique for modelling of real gas involves the use of thousands or millions of simulated molecules. The velocity components and position coordinates of the molecules are stored by the computer. These values are varied with time through representative collisions and boundary interactions in simulated space. The technique is clearly in contrast to conventional CFD method which seeks to obtain solutions of the mathematical equations that model the processes. However, due to the inability to describe rarefied gases (gases with sufficiently low density) by the Navier-Stokes equations, conventional CFD methods are incapable of handling large number of independent variables involved in the Boltzmann equation to multi-dimensional problems. This has been previously discussed in Section 1.2.2.1. Rarefied Gas Dynamics Besides that, the DSMC solution is computationally feasible because of three important factors – calculation is always unsteady with physical time being one of the important variables in simulation, effects of chemical nonequilibrium can be included by making the molecular model more complex, and there are no numerical instabilities.

The DSMC method is based on the concepts of rarefied gas physics and on the assumptions that lay the foundation for the derivation of the Boltzmann equation, but the method is not directly derived from the Boltzmann equation itself. The method is based on classical kinetic theory which causes it to be subject to the restrictions related to dilute gas and assumption of molecular chaos [40]. The method makes assumptions on three physical characteristics of a dilute gas:

1. Molecules move in random motion without interaction for time scales on the order of the local mean collision time.
2. The impact parameters and initial phase state of colliding molecules are random.
3. There are an enormous number of particles per cubic mean free path λ^3 , and only a small fraction is required for simulation in order to obtain an accurate representation of the flow.

The above-mentioned three assumptions are highly accurate for dilute gases, thus enabling the DSMC method to simulate macroscopic nonequilibrium flow fields produced by complex geometries [17].

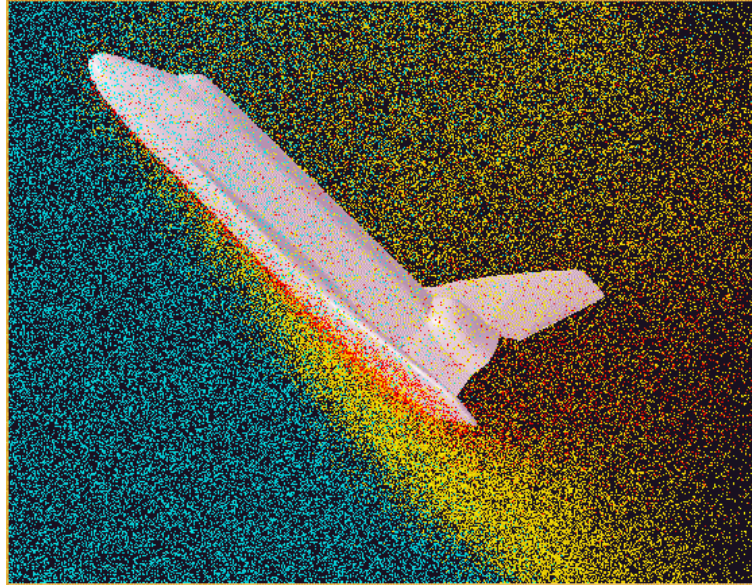


Figure 5: Simulation of the Re-Entry of the Space Shuttle using the DSMC method [41].

One of the fundamental concepts in the DSMC method is a simulated particle is not a representation of distribution of real molecules. Instead, many identical real molecules are represented in the form of one simulated molecule. This allows collisions between simulated particle pairs to be modelled with the same physical considerations of intermolecular collisions between real molecules. This is the reason due to which DSMC simulation cannot determine precisely which real molecules collide and what parameters characterise the initial conditions of each collision [17]. This is in accordance with the statement previously made stating that the DSMC method deals with intermolecular collisions using a stochastic approach rather than a deterministic approach [39].

2.1.1. Algorithm

The DSMC algorithm can be described briefly in the following way [17, 19]:

1. The simulated particles are generated at inflow boundaries
2. The particles are moved in straight lines along their velocity vectors for a timestep less than the mean collision time.
 - Boundary conditions are applied when particles collide with the surface of the geometry.
 - Particles outside the simulation domain are removed.
3. Intermolecular collisions are performed stochastically within each cell, taking into consideration the assumption that impact parameters are random in nature.
4. Particle properties in each cell are sampled.
5. Start new iteration by returning to Step 1.

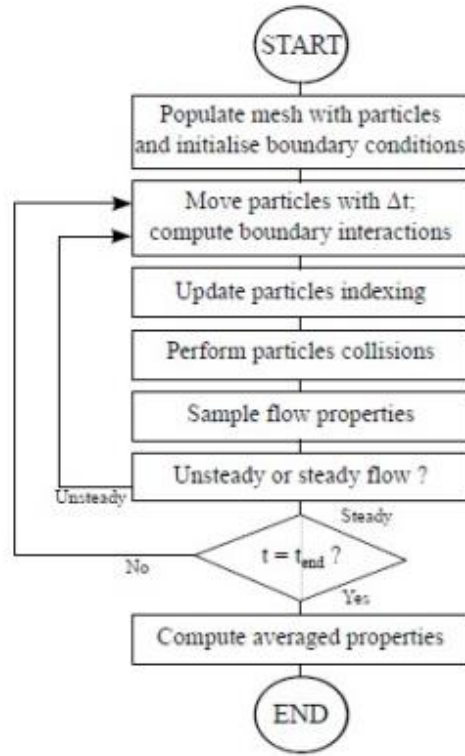


Figure 6: Algorithm Flowchart of the DSMC Method [40].

Since each simulation particle represents many real molecules in the gas, the properties of a simulated particle are the same as those of a real molecule. These properties are namely species type, their position and velocity vectors in space, and their rotational and vibrational energy. It is also useful to associate specific data with each computational cell in DSMC. The data stored in each cell can be the local particle weight, timestep, cumulative sums of samples of particle properties for individual species, volume of the cell, bounding vertices to name a few[17].

For engineering purposes, a simple molecular model is a great necessity to simulate millions or billions of molecules in a hypersonic flow simulation. The behaviour of the particles during collision depends on the choice of the intermolecular force field. The inverse power law is believed to be an accurate simple molecular model for majority of hypersonic calculations, where the inter-molecular force is modelled as the inverse of the repulsive force for a distance r between the particles [42]. The hard sphere (HS) model is a computationally effective alternative to inverse power law, where the collision cross section σ is invariant for a single gas specie, the scattering angle is isotropic and the post-collision relative speed c_r is sampled to form a uniform solid angle distribution. However, the HS molecular model is unrealistic as the total collision cross-section σ_T has no relation to the relative speed of the particles of the collision pair [23, 40]. As a result, different alternative phenomenological models have been proposed based on the HS model for accurate representation of the collisions in hypersonic flow. One of the most used, computationally efficient and reliable models is the Variable Hard Sphere (VHS) model by Bird [43].

2.1.2. Variable Hard Sphere (VHS) Model

The variable hard sphere (VHS) model defines the simulated particles as “hard spheres” and these particles have a well-defined diameter. The model follows the classical hard sphere scattering law, but the diameter is an inverse power law function of the reactive collision energy between the molecules. This is given by the expression below[43]:

$$\sigma \equiv \pi d_{ref}^2 \propto \left(\frac{1}{2} m_r c_r^2 \right)^{-\omega} \quad 24$$

where σ is the total collision cross-section, d_{ref} is the reference diameter of the particle, m_r is the reduced mass, c_r is the relative collision speed, and ω is a constant which is equal to zero. An expression for the mean free path λ in a VHS gas is required to define Knudsen numbers for the direct simulation. It is given by the expression below [44]:

$$\lambda = \left(\frac{T}{T_{ref}} \right)^{\omega} / \left[(2 - \omega)^{\omega} \Gamma(2 - \omega) 2^{\frac{1}{2}} n \sigma_{ref} \right] \quad 25$$

where T is the nonequilibrium temperature of the gas, T_{ref} is the reference free-stream temperature, n is the number density and σ_{ref} is the reference collision cross-section. The expression $(2 - \omega)^{\omega} \Gamma(2 - \omega)$ is called the “Inverse of the Variable Soft Sphere (VSS) Scattering Parameter” which is set to unity for the VHS model. From Eq.25, it can be clearly stated that the VHS model for a given gas is given by the effective diameter of the simulated particle at a given reference temperature [19].

The VHS Parameters used for the 5-species atmospheric air model used for most studies have been described in Table 2

Chemical Species	Symbol	$d_{ref} (\times 10^{-10} \text{ m})$	$T_{ref} (\text{K})$
Molecular Oxygen	O ₂	4.01	273
Molecular Nitrogen	N ₂	4.11	273
Atomic Oxygen	O	2.96	273
Atomic Nitrogen	N	2.96	273
Nitric Oxide	NO	4.14	273

Table 1: VHS Parameters for the 5 species in atmospheric air [45].

2.1.3. DS2V Solver

The DS2V solver is visual interactive implementation of Bird’s DSMC code for steady flows and unsteady flows with two independent variables. The solver is capable of simulating 2D planar and axially symmetric flows. The DSMC code used for the DS2V solver, DS2, has been modified to consider molecular fluctuations. The physical realism of these fluctuations has major consequences for DSMC calculations performed by the solver. However, the statistical scatter generally decreases with time, proportionally to the square root of the sample size. The solver employs time averaging for steady flows to attain a sufficiently small standard deviation in the results.

It must be noted that since the method's calculation is unsteady with respect to physical time, there is no convergence to the final solution, which is one of the major reasons to employ time-averaging for steady flow calculations. The DS2V program takes advantage of the major developments in the DSMC methodology since Bird's publication in 1994 [19, 46].

2.2. Atmospheric Air Model

From Section 1.2.4. Earth's Atmosphere, we can infer that the pressure, density and temperature of the atmosphere are the key parameters influencing the magnitude of the thermal stresses and aerodynamic forces acting on the spacecraft. On the other hand, the composition of the atmosphere affects the chemical interactions between the atmosphere and the surface of the spacecraft.

There are different types of air models which are used to predict gas and surface properties at hypersonic nonequilibrium conditions. The gas models used for hypersonic air flow studies vary between 5-species (neutral species), 7-species, 9-species and the 11-species models. Majority of research involving atmospheric re-entry into the Earth's atmosphere has involved the use of the 11-species model. This has resulted in development of a variety of chemical kinetic models by Park, Gupta and [47].

For the majority of DSMC calculations performed here, the atmospheric air has been modelled as a mixture of five chemical species, having three temperatures. The species are namely molecular oxygen (O_2), molecular nitrogen (N_2), atomic oxygen (O), atomic nitrogen (N) and nitric oxide (NO). The data of the model in the DS2V solver has 23 chemical reactions and active vibrational modes. Replacing the 11-species model with a 9, 7 and 5-species model significantly reduces computational resources required to model the flow. However, to obtain a numerically efficient and physically realistic representation of the hypersonic flow in DSMC, it is often necessary to know the free stream conditions up to which a certain reduced model can be used without considerable loss in accuracy.

The 5-species model has been extensively used for studies on atmospheric re-entry when the re-entry velocity of the spacecraft is near 7.5 km/s (or 7500 m/s, Mach 25)[21]. The 5-species model is a reduced version of the 11-species model. The 5-species model is obtained by not considering ionic species and electrons present in the 11-species model [29, 48, 49]. The 23 chemical reactions in the 5-species model are a combination of dissociation-recombination reactions, exchange and ionisation reactions, and bimolecular reactions.

For the reactions used to describe the chemical kinetic model of air, the rates used by researchers are typically the ones recommended by Gupta and Park [48, 50, 51]. Referring to Eq. 10, the rate constant can be rewritten as shown in Eq. 26.

$$k = A_j T^{N_j} e^{-D_j/T} \quad 26$$

Comparing Eqs.10 and 26, we get the relations for C in Eq. 10, and D_j in Eq. 26.

$$C = A_j T^{N_j}, D_j = E_a/R \quad 27$$

Figure 7 shows a table of reaction rate constants A_j , N_j and D_j .

j	Reaction	A_j	N_j	D_j
1	$O_2+O_2 \rightleftharpoons O+O+O_2$	23.243e+13	-1.0	59,500
2	$O_2+N_2 \rightleftharpoons O+O+N_2$	7.200e+12	-1.0	59,500
3	$O_2+O \rightleftharpoons O+O+O$	9.000e+13	-1.0	59,500
4	$O_2+NO \rightleftharpoons O+O+NO$	3.600e+12	-1.0	59,500
5	$O_2+N \rightleftharpoons O+O+N$	3.600e+12	-1.0	59,500
6	$N_2+O_2 \rightleftharpoons N+N+O_2$	1.900e+11	-0.5	113,500
7	$N_2+N_2 \rightleftharpoons N+N+N_2$	24.700e+11	-0.5	113,500
8	$N_2+O \rightleftharpoons N+N+O$	1.900e+11	-0.5	113,500
9	$N_2+NO \rightleftharpoons N+N+NO$	1.900e+11	-0.5	113,500
10	$N_2+N \rightleftharpoons N+N+N$	4.085e+16	-0.5	113,500
11	$NO+O_2 \rightleftharpoons N+O+O_2$	3.900e+14	-1.5	75,500
12	$NO+N_2 \rightleftharpoons N+O+N_2$	23.900e+14	-1.5	75,500
13	$NO+O \rightleftharpoons N+O+O$	27.800e+15	-1.5	75,500
14	$NO+NO \rightleftharpoons N+O+NO$	27.800e+15	-1.5	75,500
15	$NO+N \rightleftharpoons N+O+N$	27.800e+15	-1.5	75,500
16	$NO+O \rightleftharpoons N+O_2$	23.200e+03	1.0	19,700
17	$N_2+O \rightleftharpoons NO+O$	0.000e+07	0.0	38,000

Figure 7: Table showing the Reaction Rate Constants for 17 of the 23 reactions in the 5-species model of DSMC [29, 52].

The chemical composition of the 5-species model used for most studies here has been described in Table 2. It must be noted that the mole fraction numbers chosen for the chemical species to model the atmospheric air have been done in a manner such that the model is like the 5-species models used by Bird and Niu et al [21, 47].

Chemical Species	Symbol	Mole Fraction of Species
Molecular Oxygen	O_2	0.19
Molecular Nitrogen	N_2	0.75
Atomic Oxygen	O	0.02
Atomic Nitrogen	N	0.02
Nitric Oxide	NO	0.02

Table 2: Mole fractions of the chemical species in the 5-species model used for DSMC calculations in this work.

2.3. Input Parameters for DSMC Calculations in DS2V

The DS2V solver is not computationally demanding and has the following hardware requirements [53]:

- (i) Windows 98 or higher operating system
- (ii) A 2 GHz processor or higher
- (iii) 500+ megabytes (MB) of random-access memory (RAM) for efficient execution of the program
- (iv) The program gives visual output in 1024 x 768 resolution, which can be improved using post-processing software like TecPlot.

Some of the important input parameters used for setting up a DSMC calculation in the DS2V solver are as follows:

1. Flow Parameters for the Gas

- (i) The initial amount of RAM one wishes to allocate for the DSMC calculation. 80 megabytes (MB) can simulate about a million molecules.
- (ii) The type of flow field – two dimensional planar or axially symmetric.
- (iii) Reference diameter d_{ref} of the molecular model and the Inverse of the VSS scattering parameter, which is set to 1 to use the VHS model.
- (iv) The atmospheric air/gas model
- (v) Temperature of the free stream reference gas.
- (vi) Mole fractions of the species in the air model.
- (vii) The type of calculation a user wishes to perform – steady flow at large times or continuous unsteady flow.
- (viii) Number Density of the free stream reference gas. The number density is dependent on the pressure and the temperature of the gas, and the value can be obtained from Eq. 28, as shown below:

$$n = p/kT \quad 28$$

where n is the number density, p is the pressure, T is the temperature and k is the Boltzmann constant which is equal to 1.380649×10^{-23} J/K

2. Surface and Geometry Parameters

- (i) Surface definition – whether the surface is open or closed.
- (ii) Number of segments that the surface has been divided into to specify its geometry.
- (iii) The initial and final point of each segment. In our case, all our segments are straight lines. However, segments can be defined as arcs as well.
- (iv) Number of sampling intervals along the segment.
- (v) The surface temperature for molecular reflection.
- (vi) Emissivity of the surface. The emissivity of the surface should be the emissivity of the material used to design the structure of the spacecraft.

- (vii) The reflection model one wishes to use to model particle interactions with the surface – diffuse reflection or Cercignani-Lampis-Lord (CLL) reflection model [54].
- (viii) Fraction of specular reflection and adsorption. Full specular reflection can be used to make an inviscid flow calculation. The specified fraction of adsorption removes a certain number of molecules that strike the surface. Typically, the fractions add up to unity.

3. Boundary Conditions

- (i) The coordinates of the minimum and maximum x-boundaries of the planar or axially symmetric flow field.
- (ii) The coordinates of the minimum and maximum y-boundaries of the planar or axially symmetric flow field.
- (iii) The type of interface the boundary has with the initial flow and the initial state of the flow field. The interface can either be set to the stream or reference gas, or vacuum.

Most of the studies presented in this work have been given the following values to define the hypersonic flow representation:

Parameter (unit)	Defined Value
Atmospheric air model for reference gas	Real air 5-species model
Mole fraction of the chemical species	Refer to Table 2
Reference diameter of molecules	Refer to Table 1
Type of flow field	Two-dimensional planar
Number density (per cubic metre)	3.498×10^{20}
Temperature (K)	203.18 (unless stated)
Inverse of VSS Scattering Parameter	1 (to use VHS model)
Initial RAM Allocation (MB)	200
Type of solution	Steady flow at large times

Table 3: Defined values for the flow parameters of the gas in DS2V for studies presented in this work.

Parameter (unit)	Defined Value
Minimum x-coordinate (m)	-0.5
Maximum x-coordinate (m)	0.5
Minimum y-coordinate (m)	-0.5
Maximum y-coordinate (m)	0.5
Type of interface with boundary	Stream or reference gas
Initial state of the flow field	Primary stream

Table 4: Defined values for the boundary condition parameters in DS2V for studies presented in this work.

The defined values for the surface and geometry parameters have been discussed in greater detail in Section 2.4. **Surface Modelling and Geometry**

Parameters because it relates to one of the parameters under investigation – the CubeSat geometry, which has been discussed in Section 4.2.3. Geometry of CubeSat

2.4. Surface Modelling and Geometry Parameters

The surface of the geometries under investigation has been modelled as an insulated adiabatic body with a surface temperature of 2000 K and emissivity of 0.03. These values have been taken because of thermal insulation paints like Solar White can generate surface temperatures around 2000 K while insulating the payload and the components of the CubeSat [55]. Thermal paints also allow the components to be within their operational temperature range which is typically from -40°C to 90°C . The emissivity of the surface is the material property of Al-6061 aluminium alloy. We have opted to choose the diffuse reflection model as the model can provide an accurate representation of interactions between the molecules and the surface at re-entry altitudes of 80 to 90 km and re-entry velocity of 6.5 to 7.1 km/s. To model viscous flow, the fraction of specular reflection has been varied between the ranges of 0.9 to 0.8, which results in the fraction of adsorption to range between 0.1 to 0.2, as the two fraction values sum up to unity[52].

The geometry parameters used for the DSMC calculations in the DS2V solver vary according to the geometry of the CubeSat along which we wish to simulate the hypersonic flow. Each CubeSat geometry has been modelled as one surface with 4 straight line segments. For CubeSat configurations greater than 1U i.e., 2U, 3U and 6U CubeSats, the geometry parameters can be slightly altered to obtain different orientations. Table 5 tabulates the geometry parameters for different CubeSats.

CubeSat	Dimensions (cm)	End Point S1	End Point S2	End Point S3	End Point S4
1U	$10 \times 10 \times 10$	(0.05, -0.05)	(0.05, 0.05)	(-0.05, 0.05)	(-0.05, -0.05)
2U	$20 \times 10 \times 10$	(0.15, -0.05)	(0.15, 0.05)	(-0.05, 0.05)	(-0.05, -0.05)
3U	$30 \times 10 \times 10$	(0.25, -0.05)	(0.25, 0.05)	(-0.05, 0.05)	(-0.05, -0.05)
6U	$34 \times 20 \times 10$	(0.31, -0.05)	(0.31, 0.18)	(-0.05, 0.18)	(-0.05, -0.05)

Table 5: Geometry parameters of different CubeSat configurations in DS2V, with start point of the surface fixed at $(-0.05, -0.05)$.

2.5. Output Files and Postprocessing

After solving for the hypersonic flow, the DS2 program generates ASCII files DS2FF.DAT and DS2FFxx.DAT, where xx is the two digit species number when a gas mixture model like the ones described in Section 2.2. **Atmospheric Air Model** is used to model the reference gas. The DS2 program may generate the ASCII file DS2SU.DAT which contain the output for the macroscopic surface

properties. The .DAT files can be used to obtain temperature and density distribution results using the

The descriptive column headings for the surface properties in the flow field is shown in Table 6:

Column Heading	Description
S	Distance along surface from initial point to centre of interval
X	X-coordinate of the centre of the interval
Y	Y-coordinate of the centre of the interval
NF	Number flux to the surface
PR	Surface pressure
SSP	Shear Stress in the plane of the flow
SSC	Shear Stress normal to the flow
EI	Incident energy flux
ER	Reflected energy flux
TS	Net energy flux
VG	Velocity Slip
TG	Translational temperature slip
TRG	Rotational temperature slip
Ni	The sample of species i

Table 6: Output Data from the DS2SU.DAT file generated by the DS2V solver[53].

Table 7 tabulates the descriptive column headings for the flow mixture flow field:

Column Heading	Description
X	X-coordinate
Y	Y-coordinate
ND	Number Density
D	Density
TTR	Translational Temperature
TRT	Rotational Temperature
TVB	Vibrational Temperature
TOV	Overall Temperature
MA	Mach Number
P	Scalar pressure

Table 7: Output Data from the DS2FF.DAT file generated by the DS2V solver[53].

The descriptive column headings for the single species in the flow mixture flow field properties have been tabulated in Table 8:

Column Heading	Description
X	X-coordinate
Y	Y-coordinate
SAMP	Sample size
TTR	Translational Temperature
TRT	Rotational Temperature

TVB

Vibrational Temperature

TOV

Overall Temperature

Table 8: Output Data from the DS2FFxx.DAT file generated by the DS2V solver[53].

3. Results

The results illustrated in Section 3.1. **Angle of Attack**, 3.2. **Solar Activity**, 3.3. **Geometry of the CubeSat**, and 3.4. **Atmospheric Gas Models** consists of density and temperature distributions in a 2-dimensional planar flow field based on the 3-T model. The units for output parameters D is kg/m^3 , kelvin (K) for temperatures TTR, TRT, TVB. The unit for flow field parameters X and Y is metre (m).

3.1. Angle of Attack

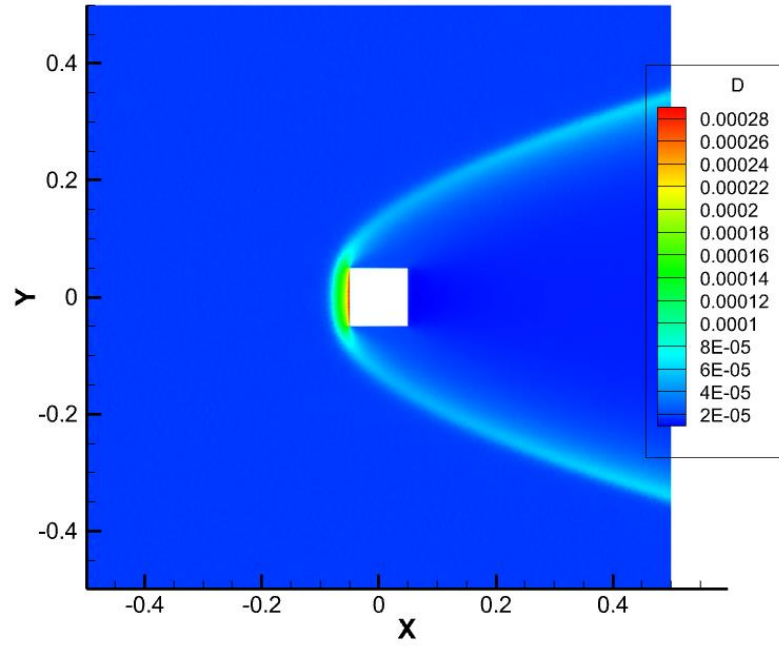


Figure 8: Density Distribution of Mach 25 flow across 1U CubeSat at 0° angle of attack

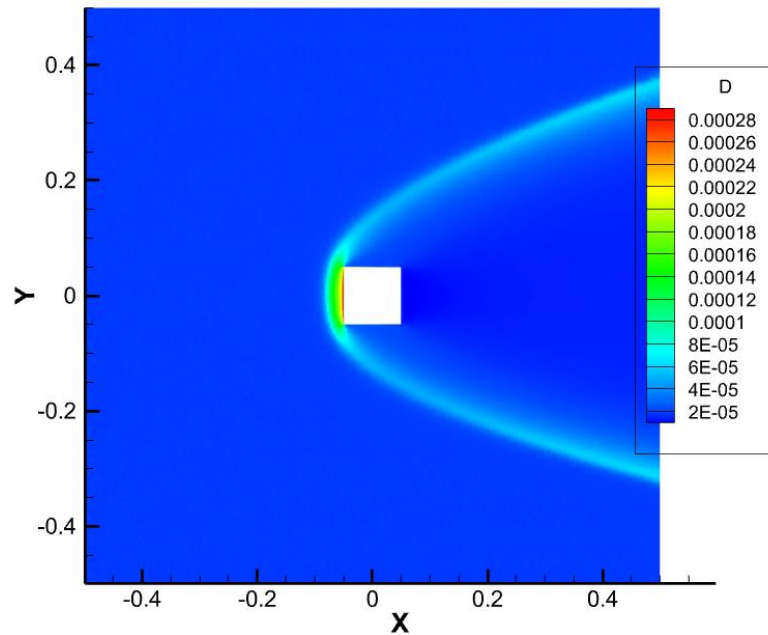


Figure 9: Density Distribution of Mach 25 flow across 1U CubeSat at 2.5° angle of attack

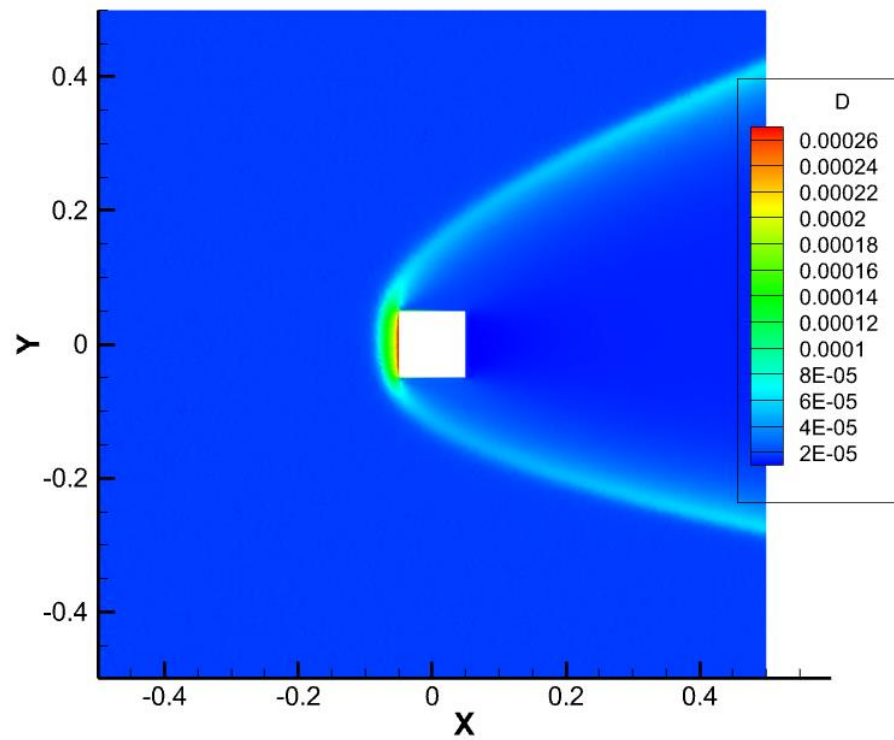


Figure 10: Density Distribution of Mach 25 flow across 1U CubeSat at 0° angle of attack

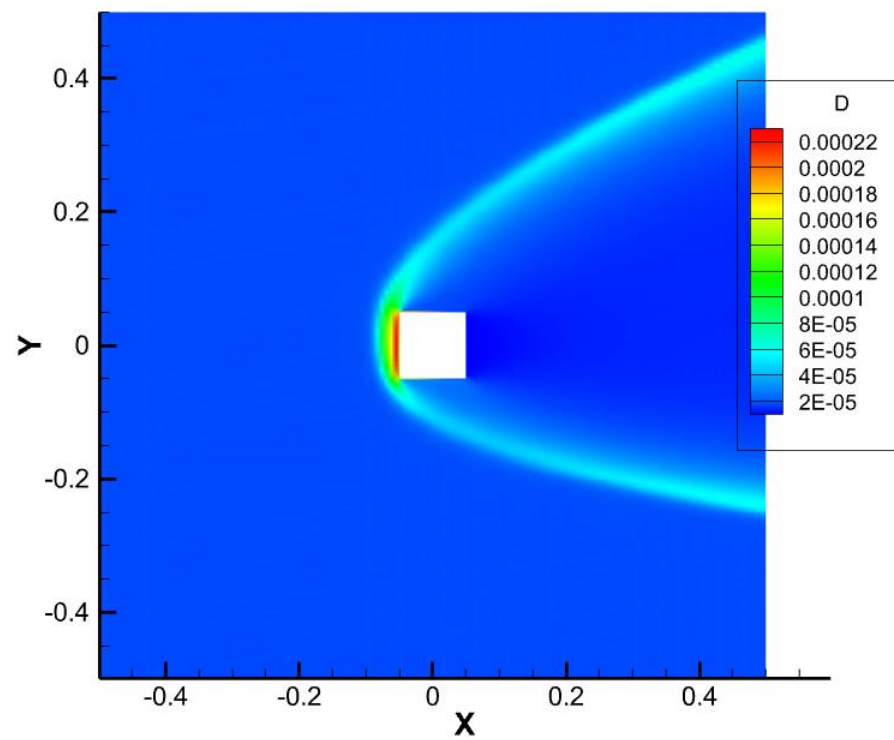


Figure 11: Density Distribution of Mach 25 flow across 1U CubeSat at 7.5° angle of attack

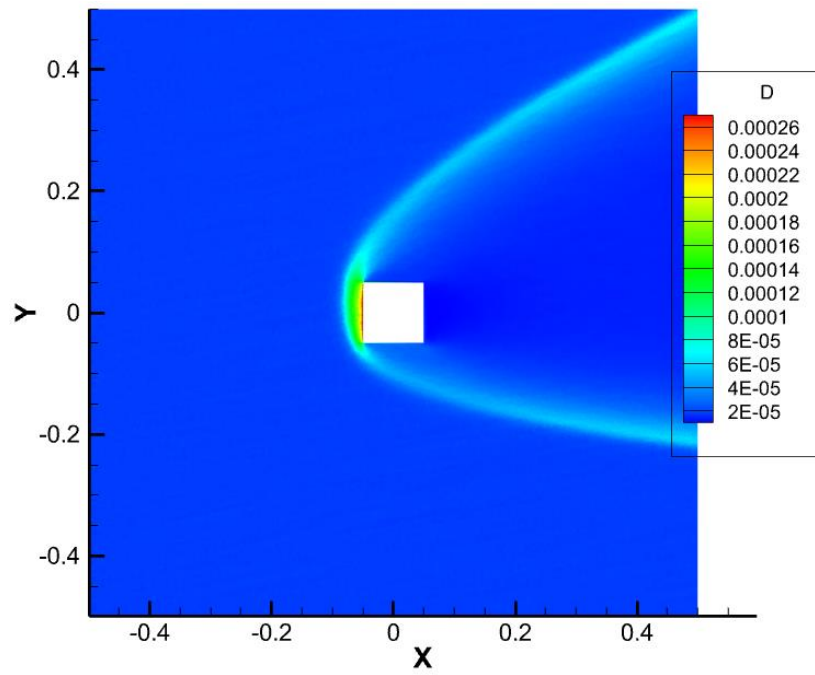


Figure 12: Density Distribution of Mach 25 flow across 1U CubeSat at 10° angle of attack

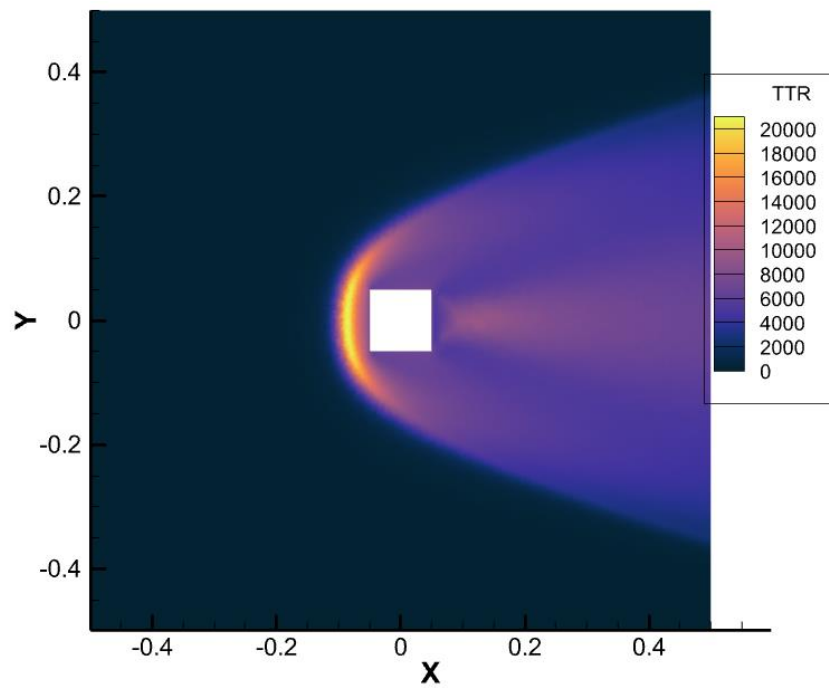


Figure 13: Translational Temperature Distribution of Mach 25 flow across 1U CubeSat at 0° angle of attack

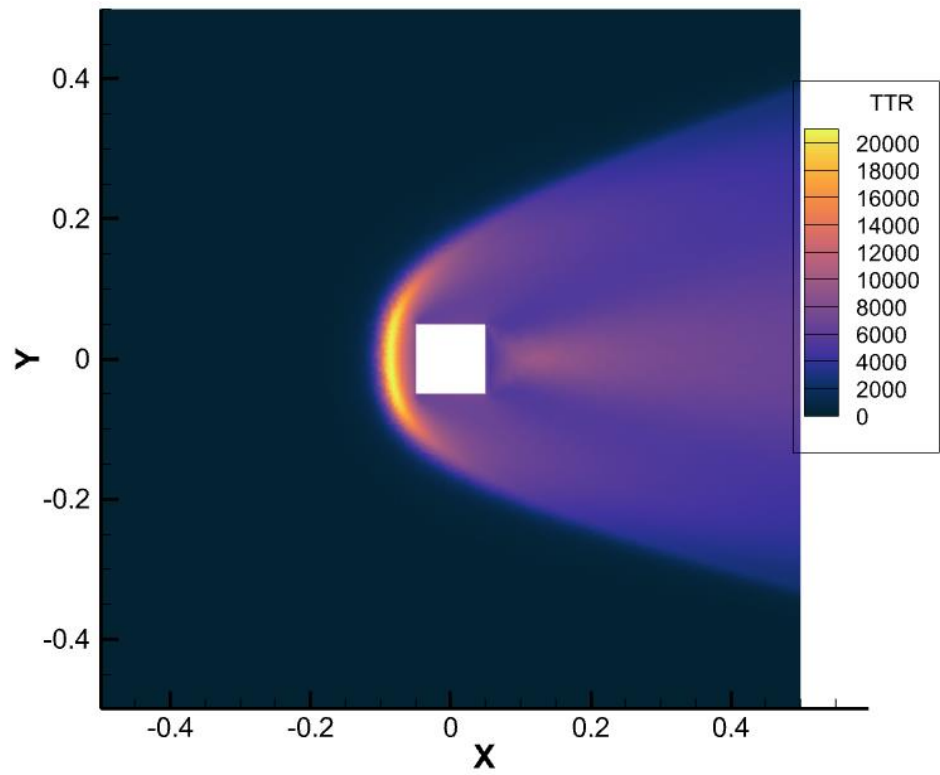


Figure 14: Translational Temperature Distribution of Mach 25 flow across 1U CubeSat at 2.5° angle of attack

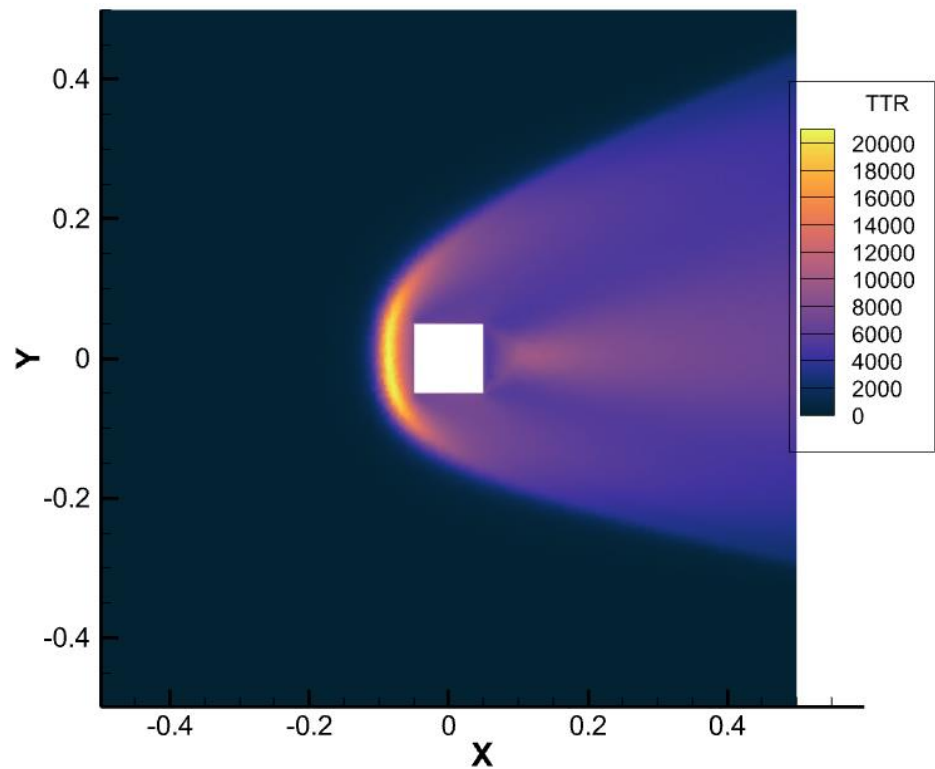


Figure 15: Translational Temperature Distribution of Mach 25 flow across 1U CubeSat at 5° angle of attack

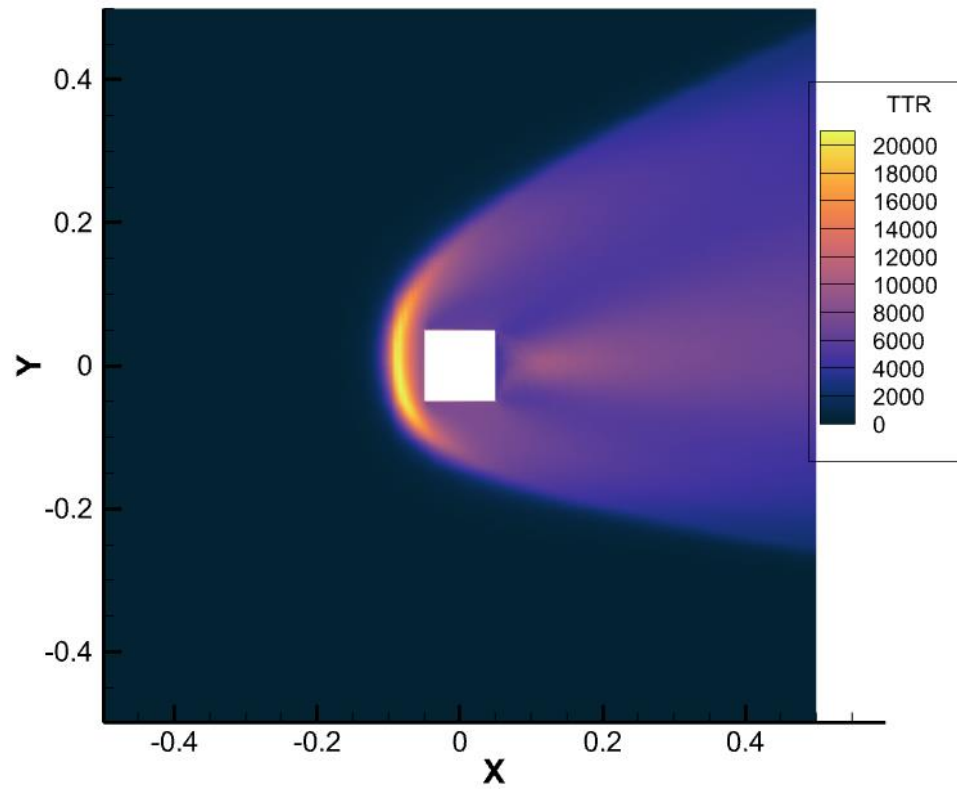


Figure 16: Translational Temperature Distribution of Mach 25 flow across 1U CubeSat at 7.5° angle of attack

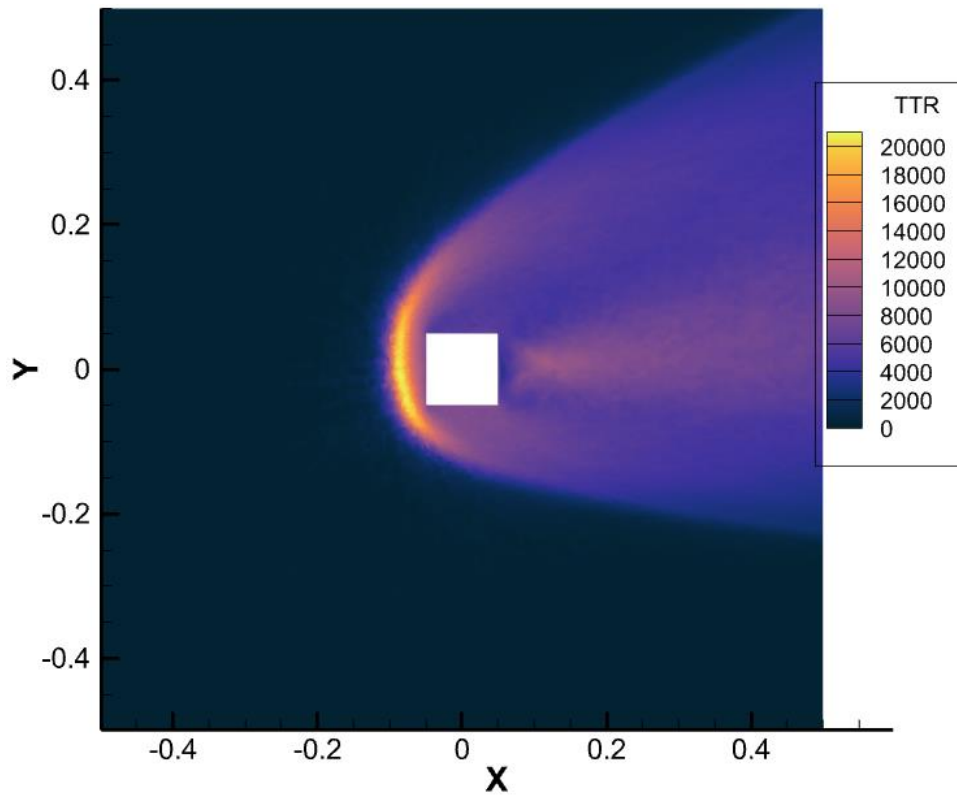


Figure 17: Translational Temperature Distribution of Mach 25 flow across 1U CubeSat at 10° angle of attack

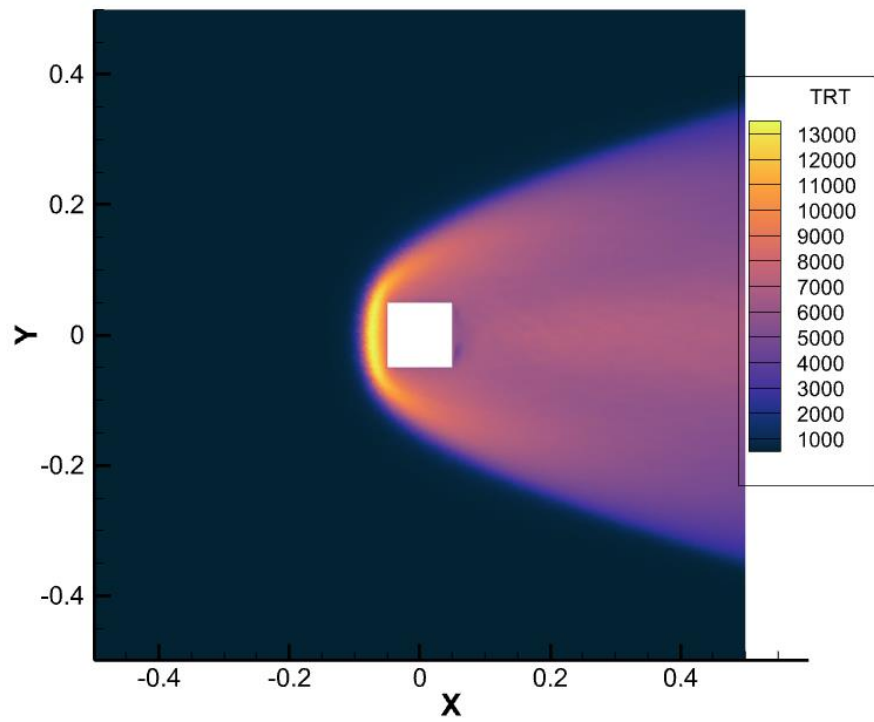


Figure 18: Rotational Temperature Distribution of Mach 25 flow across 1U CubeSat at 0° angle of attack

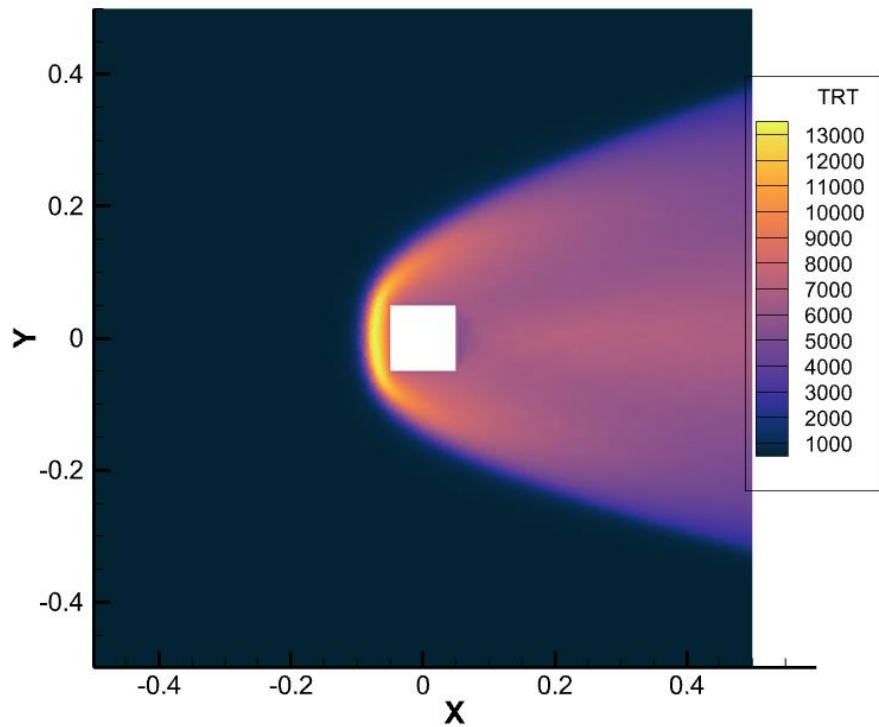


Figure 19: Rotational Temperature Distribution of Mach 25 flow across 1U CubeSat at 2.5° angle of attack

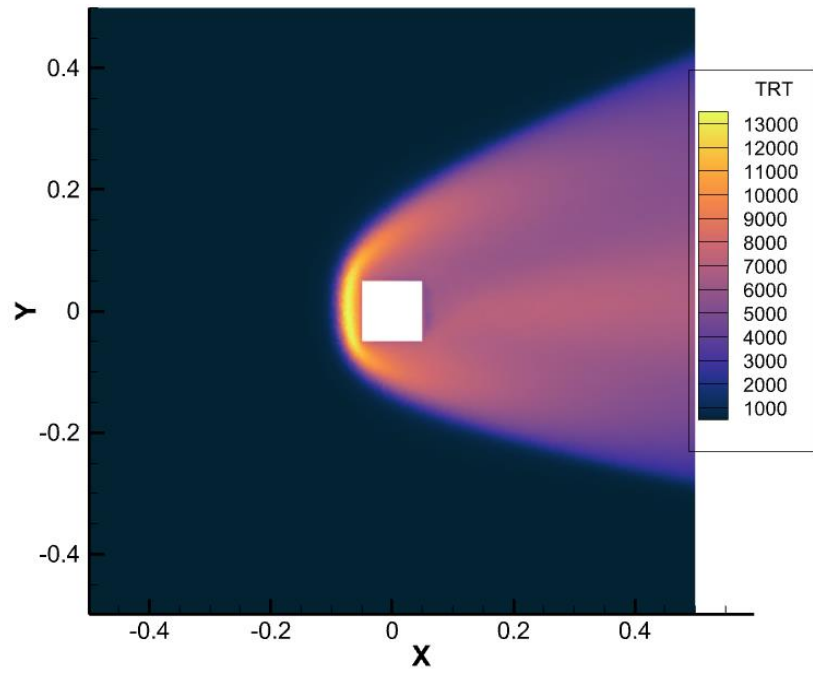


Figure 20: Rotational Temperature Distribution of Mach 25 flow across 1U CubeSat at 5° angle of attack

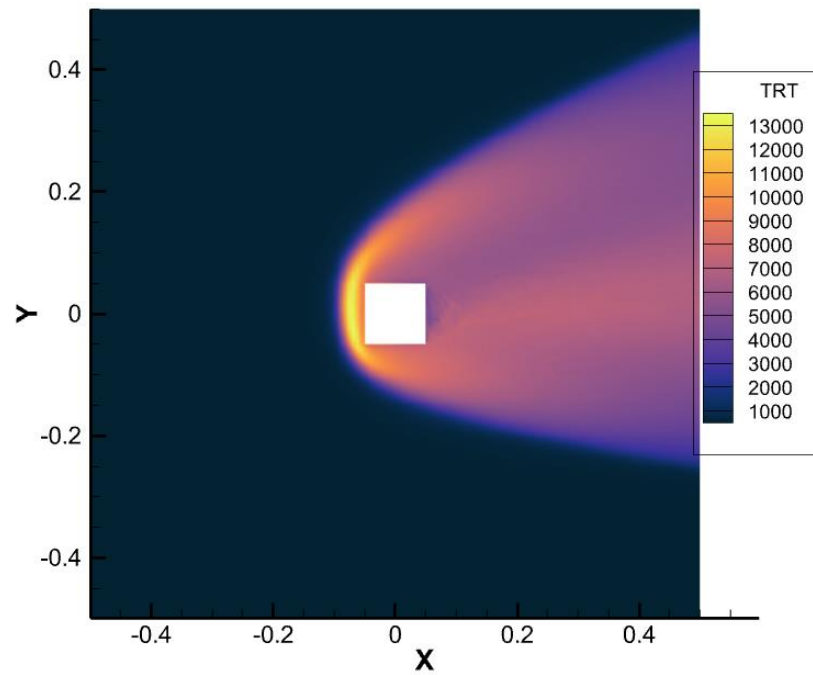


Figure 21: Rotational Temperature Distribution of Mach 25 flow across 1U CubeSat at 7.5° angle of attack

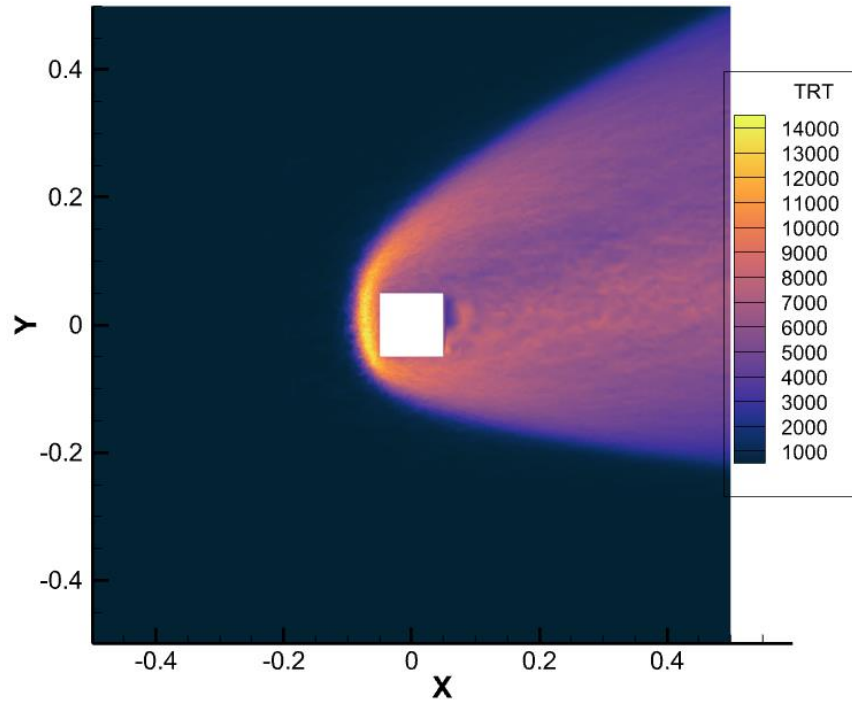


Figure 22: Rotational Temperature Distribution of Mach 25 flow across 1U CubeSat at 10° angle of attack

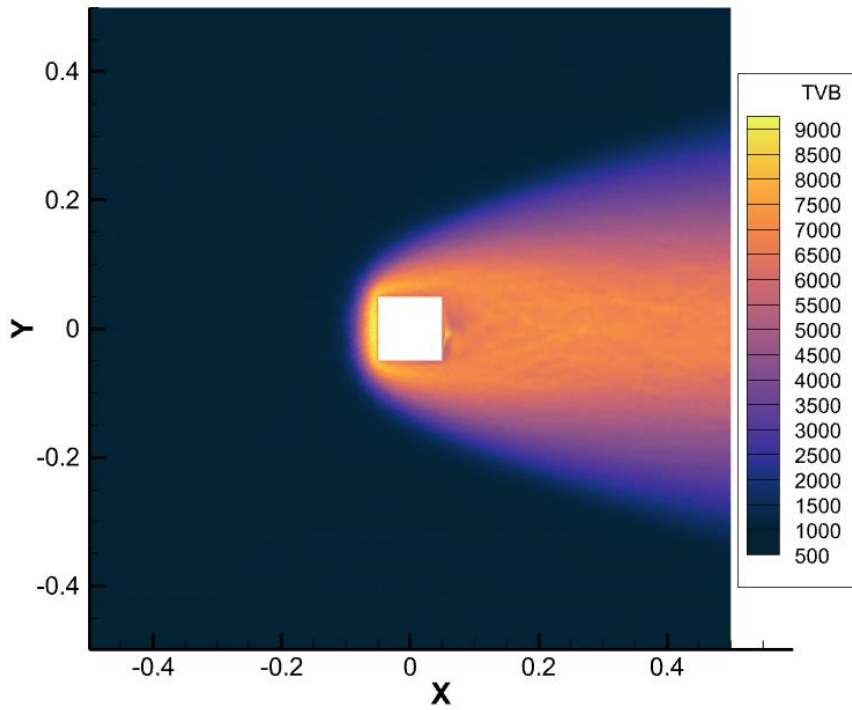


Figure 23: Vibrational Temperature Distribution of Mach 25 flow across 1U CubeSat at 0° angle of attack

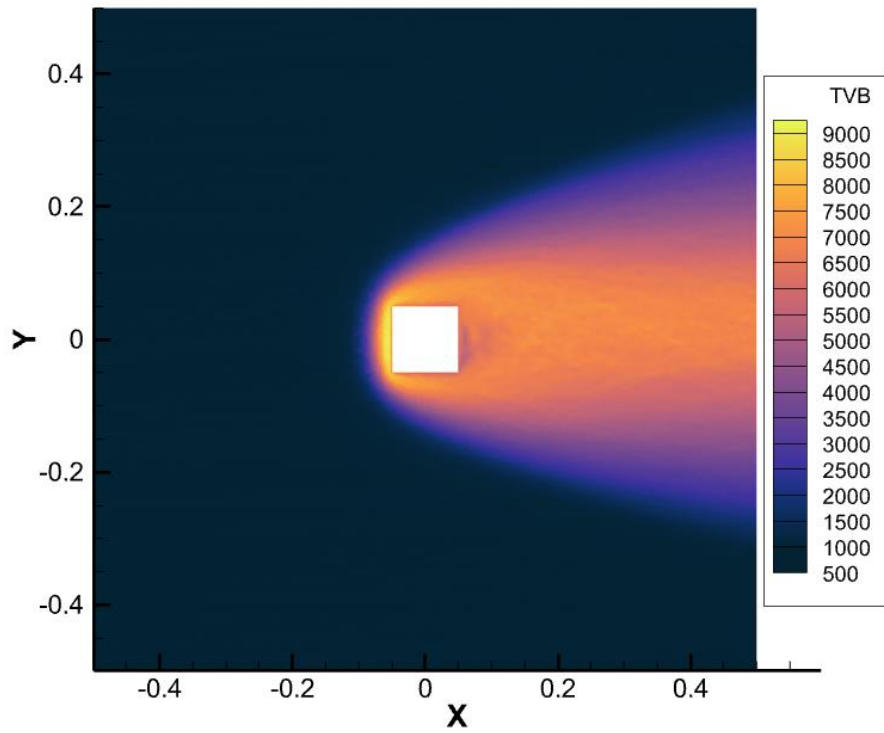


Figure 24: Vibrational Temperature Distribution of Mach 25 flow across 1U CubeSat at 2.5° angle of attack

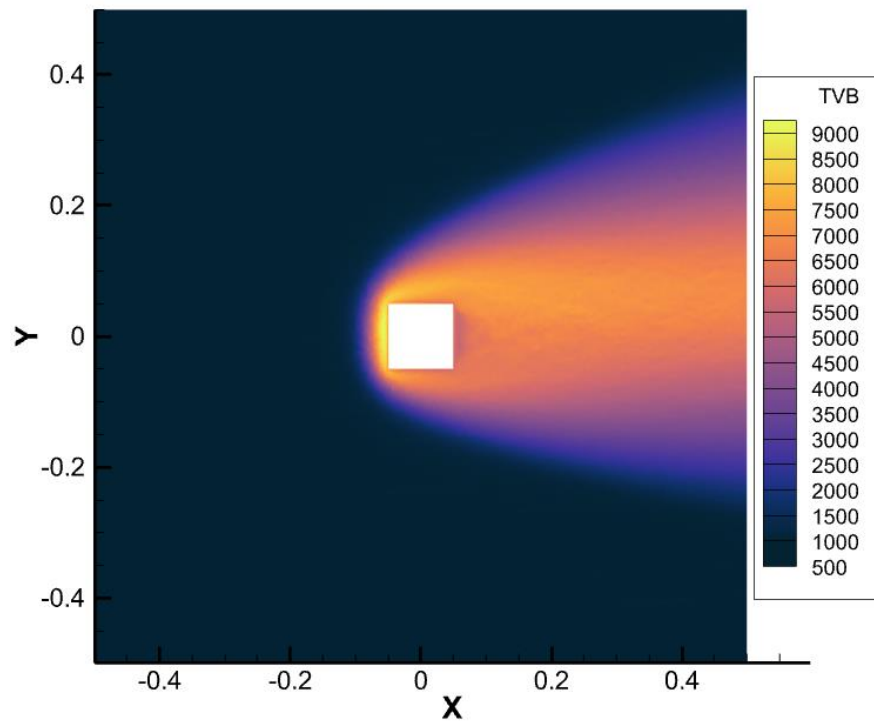


Figure 25: Vibrational Temperature Distribution of Mach 25 flow across 1U CubeSat at 5° angle of attack

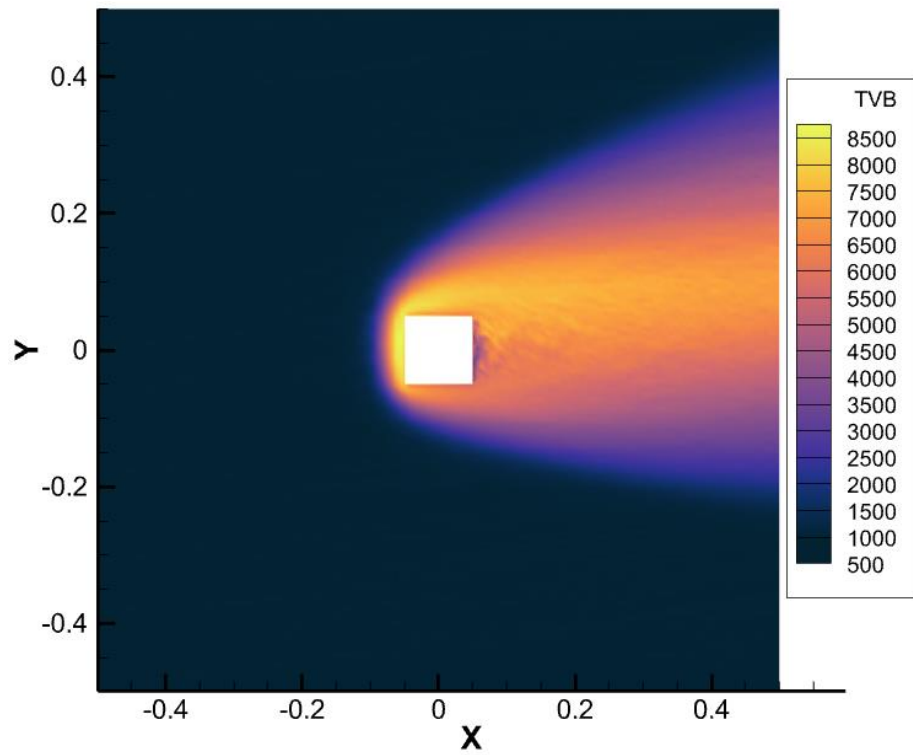


Figure 26: Vibrational Temperature Distribution of Mach 25 flow across 1U CubeSat at 7.5° angle of attack

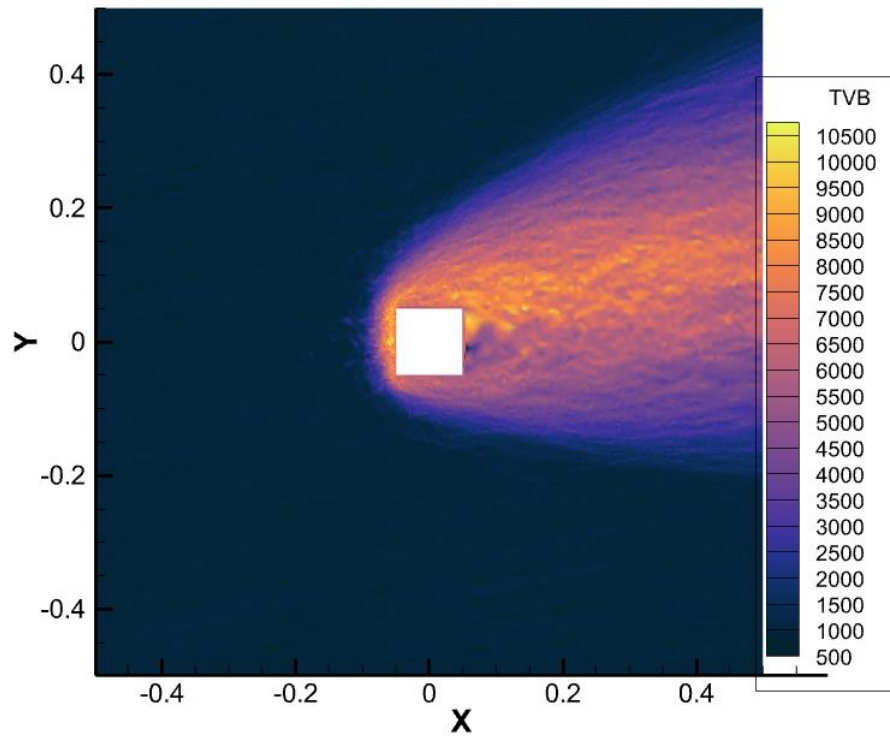


Figure 27: Vibrational Temperature Distribution of Mach 25 flow across 1U CubeSat at 10° angle of attack

3.2. Solar Activity

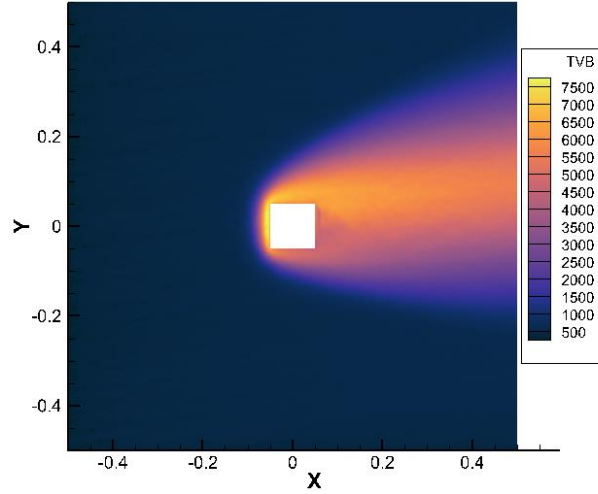


Figure 28: Vibrational Temperature Distribution of Mach 25 flow across 1U CubeSat at 7.5° angle of attack and Low Solar Activity

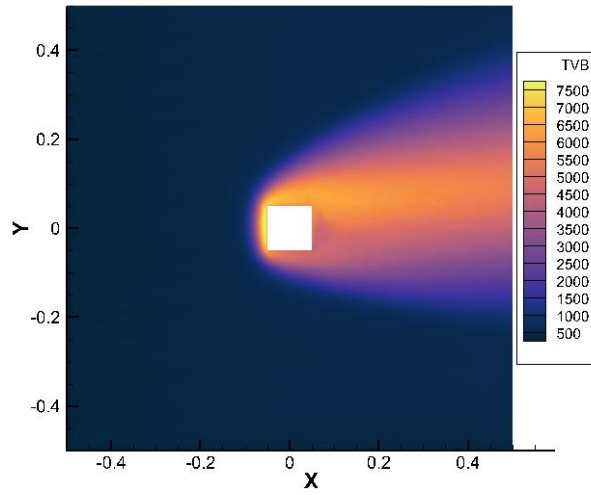


Figure 29: Vibrational Temperature Distribution of Mach 25 flow across 1U CubeSat at 7.5° angle of attack and Mean Solar Activity

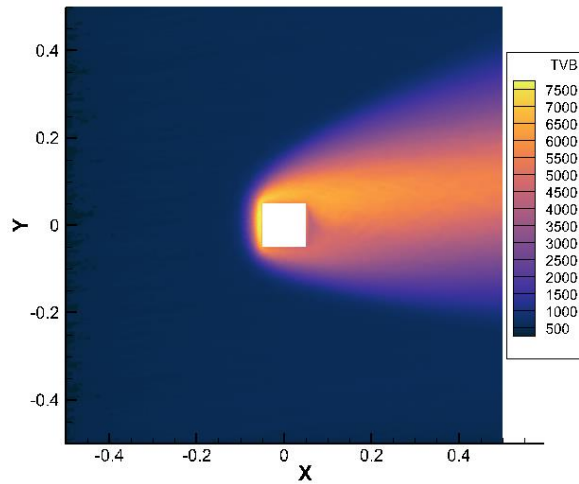


Figure 30: Vibrational Temperature Distribution of Mach 25 flow across 1U CubeSat at 7.5° angle of attack and Extremely High Solar Activity

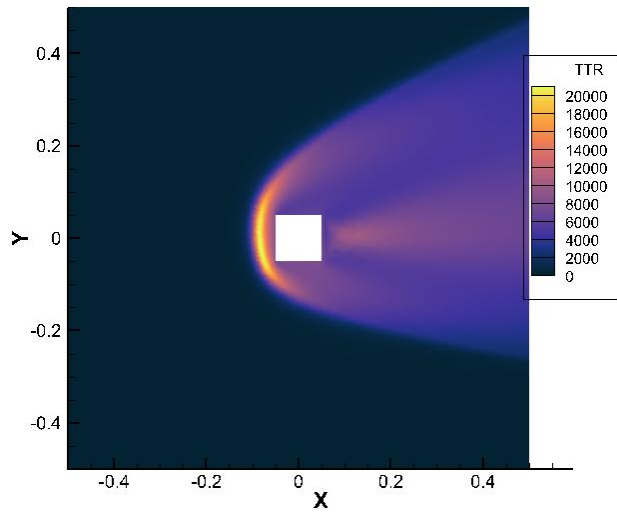


Figure 31: Translational Temperature Distribution of Mach 25 flow across 1U CubeSat at 7.5° angle of attack and Low Solar Activity

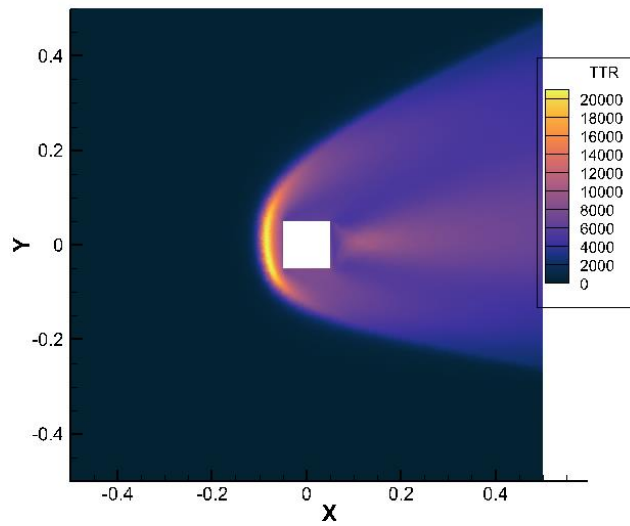


Figure 32: Translational Temperature Distribution of Mach 25 flow across 1U CubeSat at 7.5° angle of attack and Mean Solar Activity

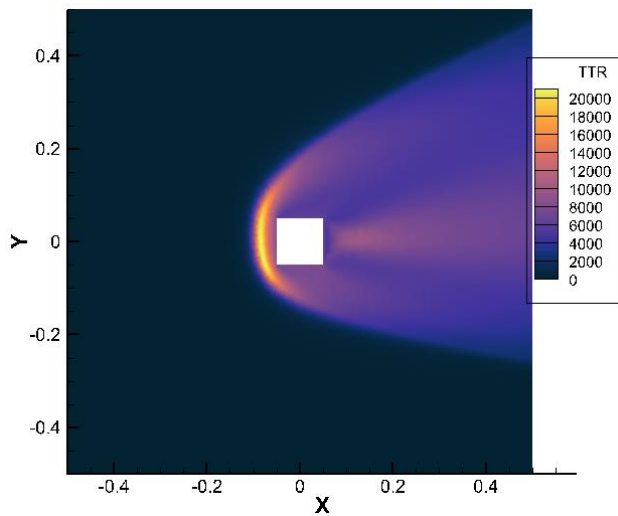


Figure 33: Translational Temperature Distribution of Mach 25 flow across 1U CubeSat at 7.5° angle of attack and Extremely High Solar Activity

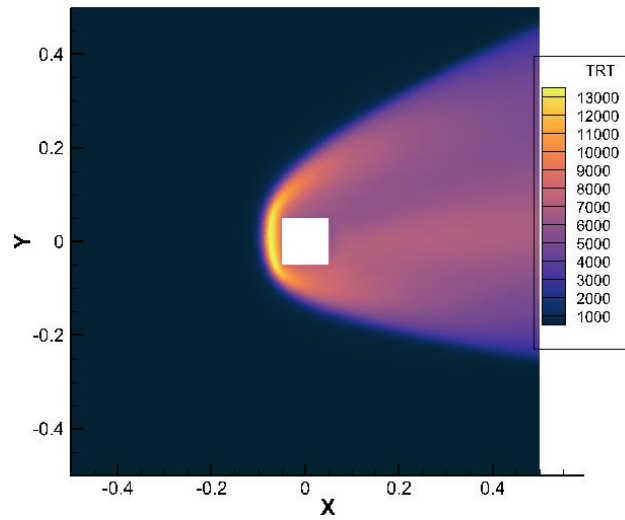


Figure 34: Rotational Temperature Distribution of Mach 25 flow across 1U CubeSat at 7.5° angle of attack and Low Solar Activity

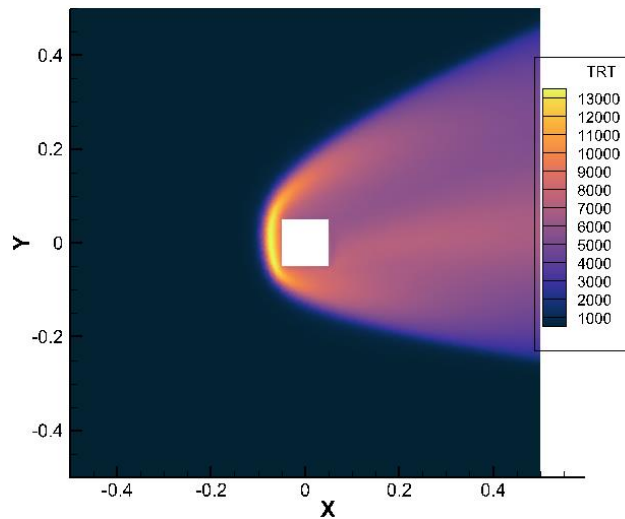


Figure 35: Rotational Temperature Distribution of Mach 25 flow across 1U CubeSat at 7.5° angle of attack and Mean Solar Activity

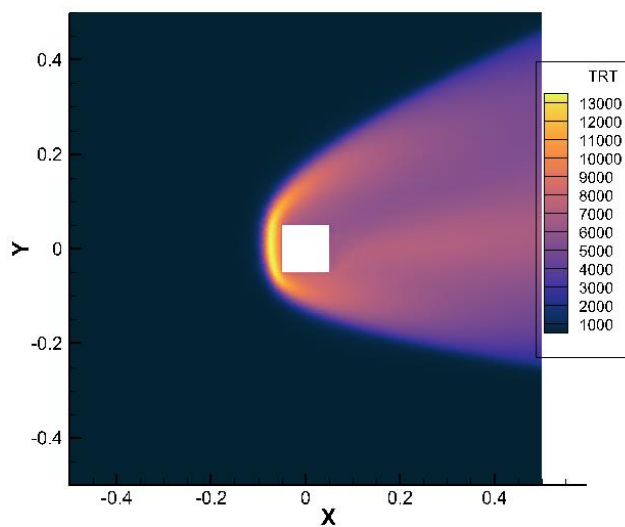


Figure 36: Rotational Temperature Distribution of Mach 25 flow across 1U CubeSat at 7.5° angle of attack and Extremely High Solar Activity

3.3. Geometry of the CubeSat

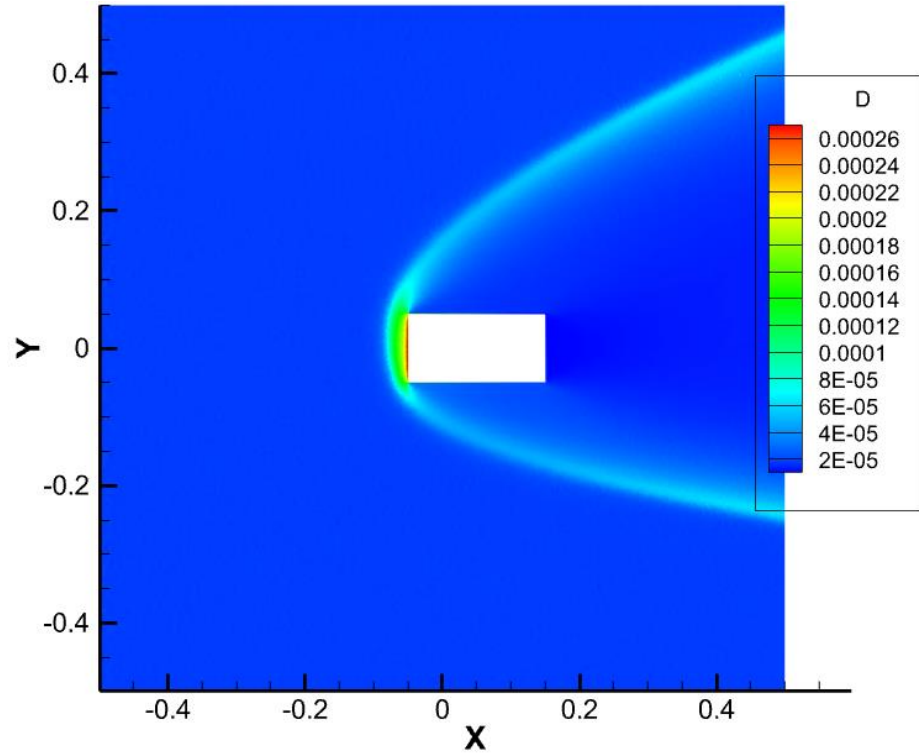


Figure 37: Density Distribution of Mach 25 flow across 2U CubeSat, Side +Z face-pointing at 7.5° angle of attack

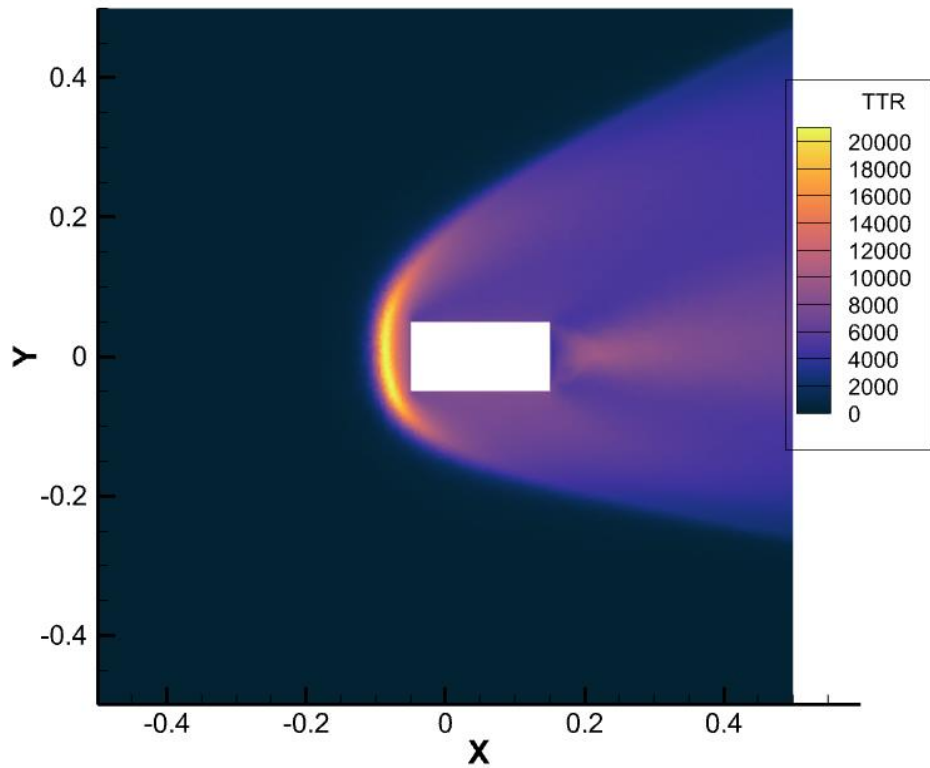


Figure 38: Translational Temperature Distribution of Mach 25 flow across 2U CubeSat, Side +Z face-pointing at 7.5° angle of attack

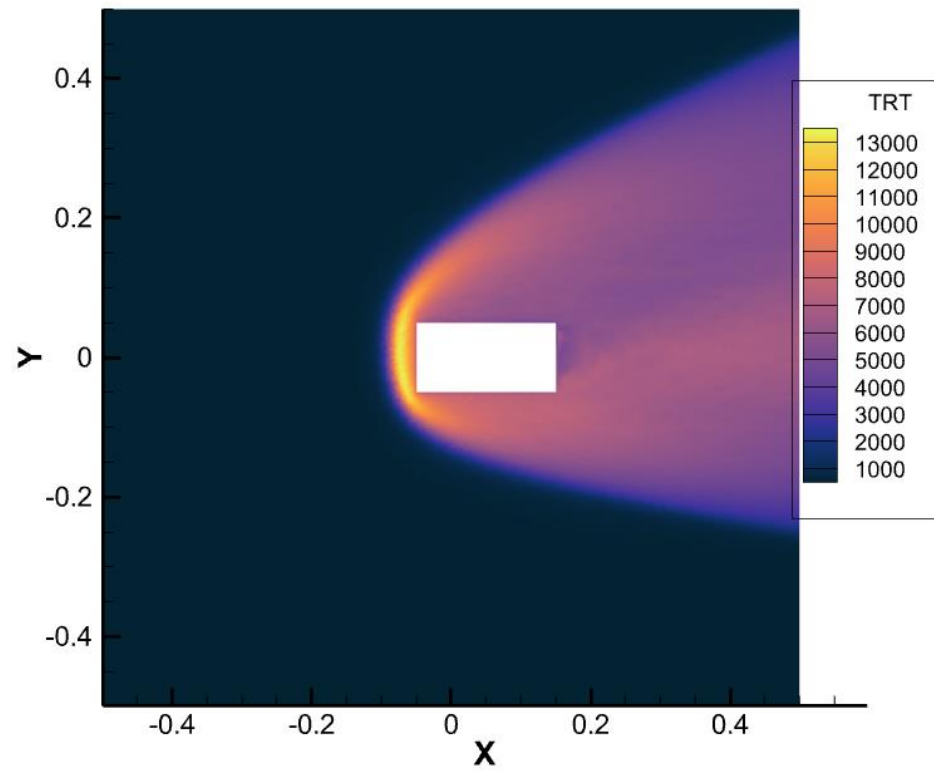


Figure 39: Rotational Temperature Distribution of Mach 25 flow across 2U CubeSat, Side +Z face-pointing at 7.5° angle of attack

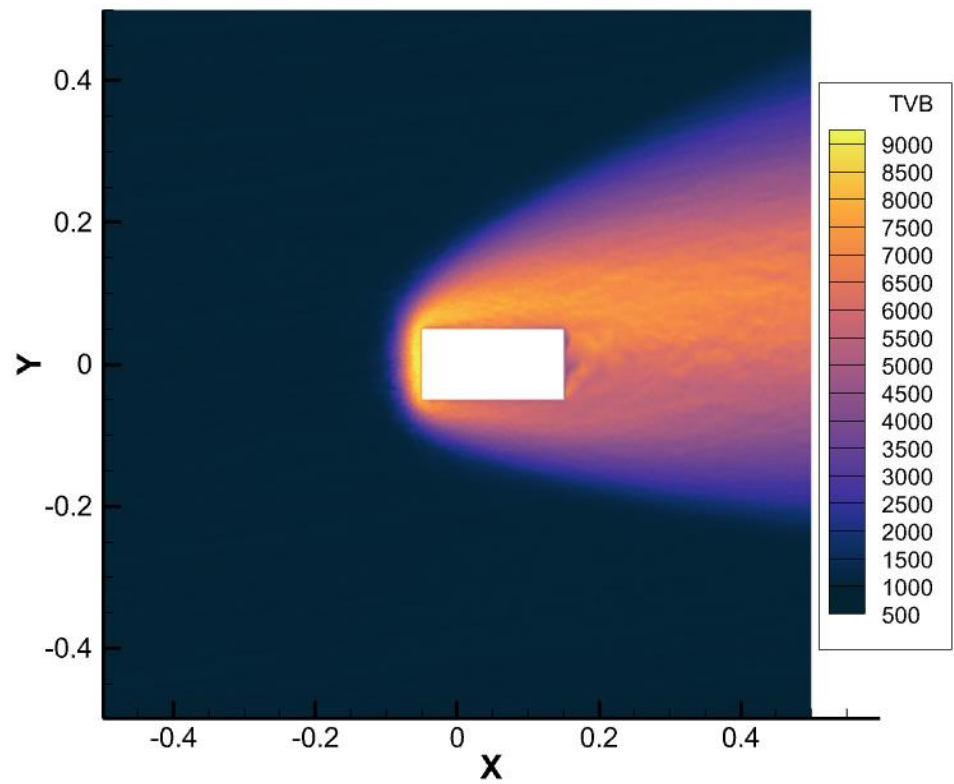


Figure 40: Vibrational Temperature Distribution of Mach 25 flow across 2U CubeSat, Side +Z face-pointing at 7.5° angle of attack

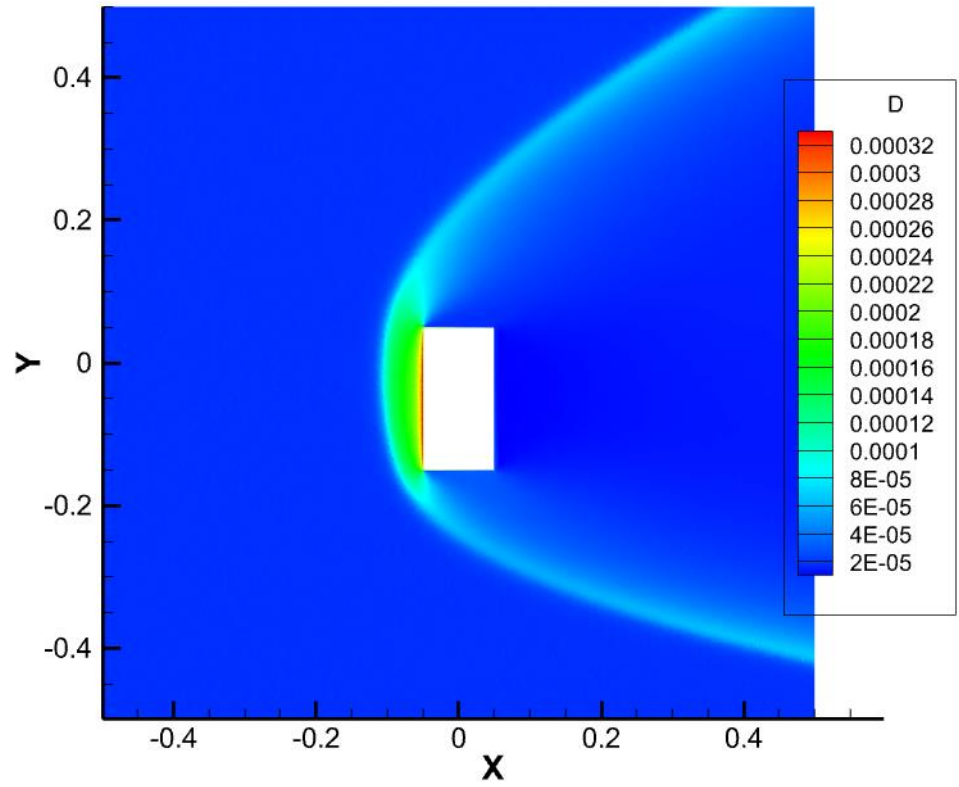


Figure 41: Density Distribution of Mach 25 flow across 2U CubeSat, Side +X or +Y face-pointing, at 7.5° angle of attack

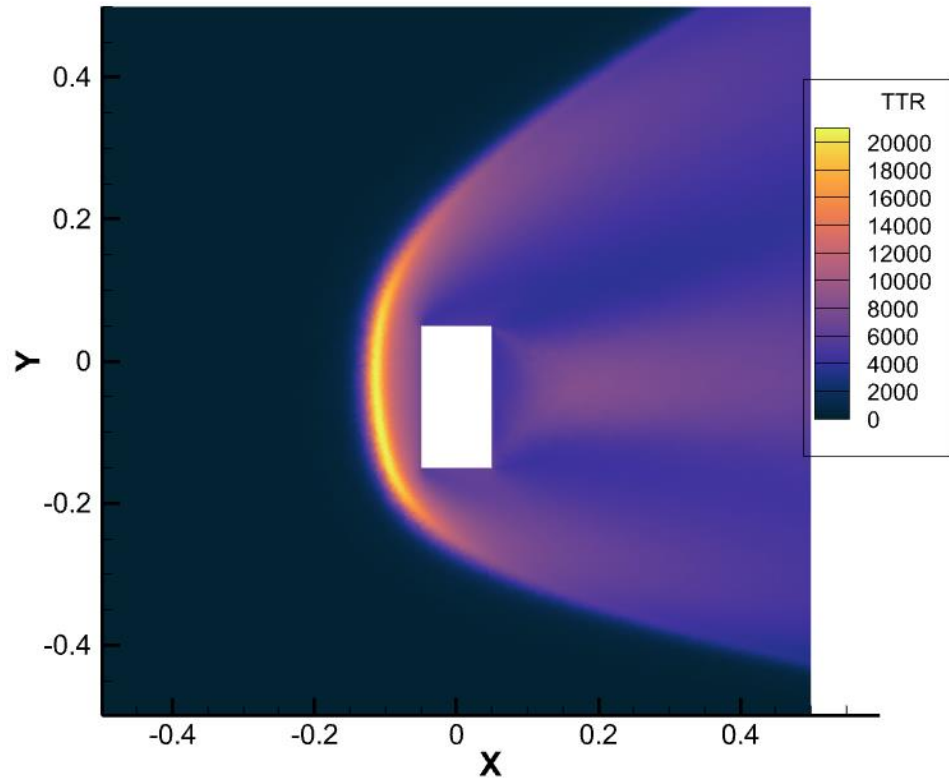


Figure 42: Translational Temperature Distribution of Mach 25 flow across 2U CubeSat, Side +X or +Y face-pointing, at 7.5° angle of attack

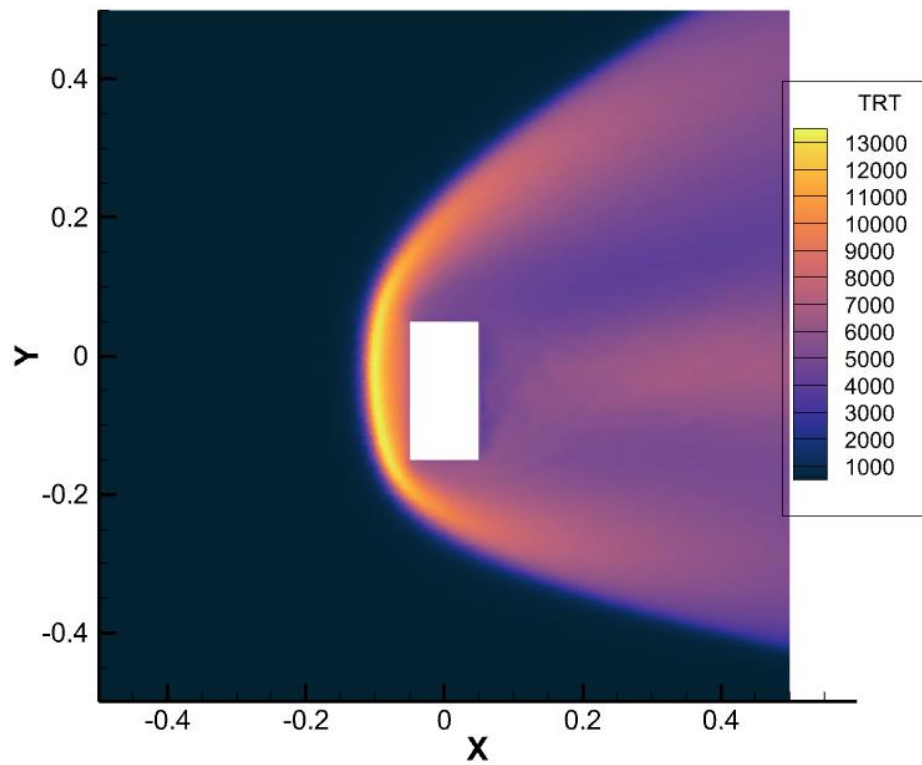


Figure 43: Rotational Temperature Distribution of Mach 25 flow across 2U CubeSat, Side +X or +Y face-pointing at 7.5° angle of attack

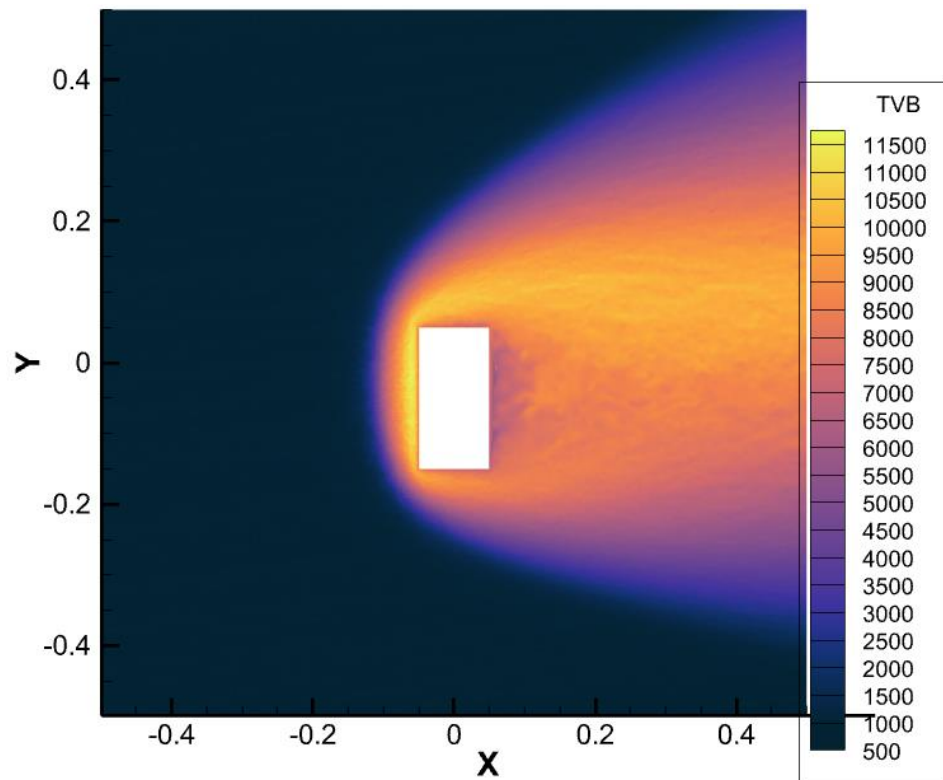


Figure 44: Vibrational Temperature Distribution of Mach 25 flow across 2U CubeSat, Side +X or +Y face-pointing at 7.5° angle of attack

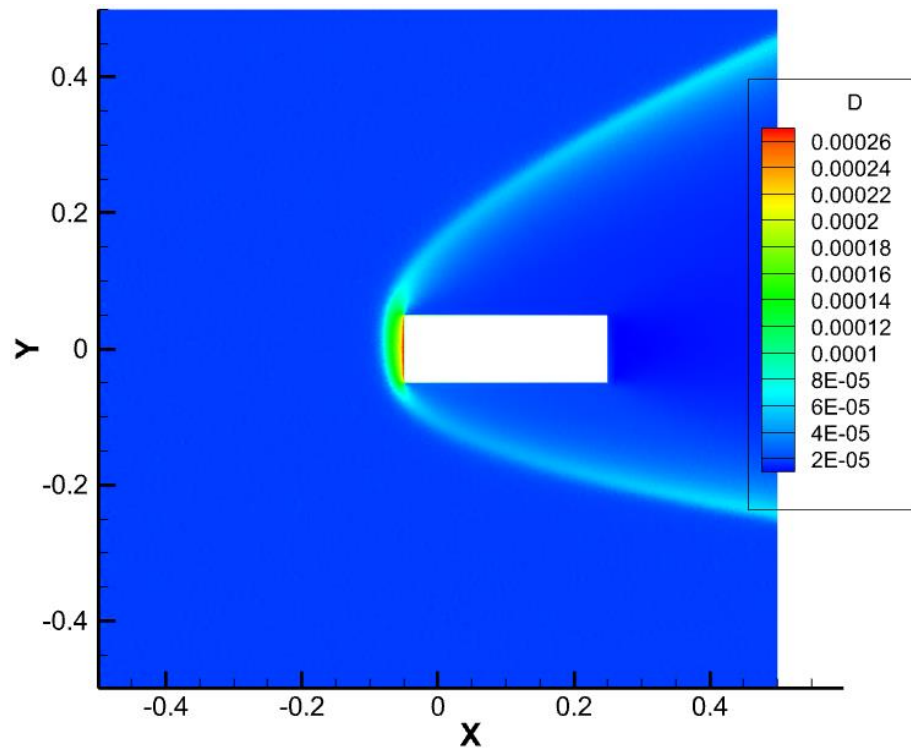


Figure 45: Density Distribution of Mach 25 flow across 3U CubeSat, Side +Z face-pointing at 7.5° angle of attack

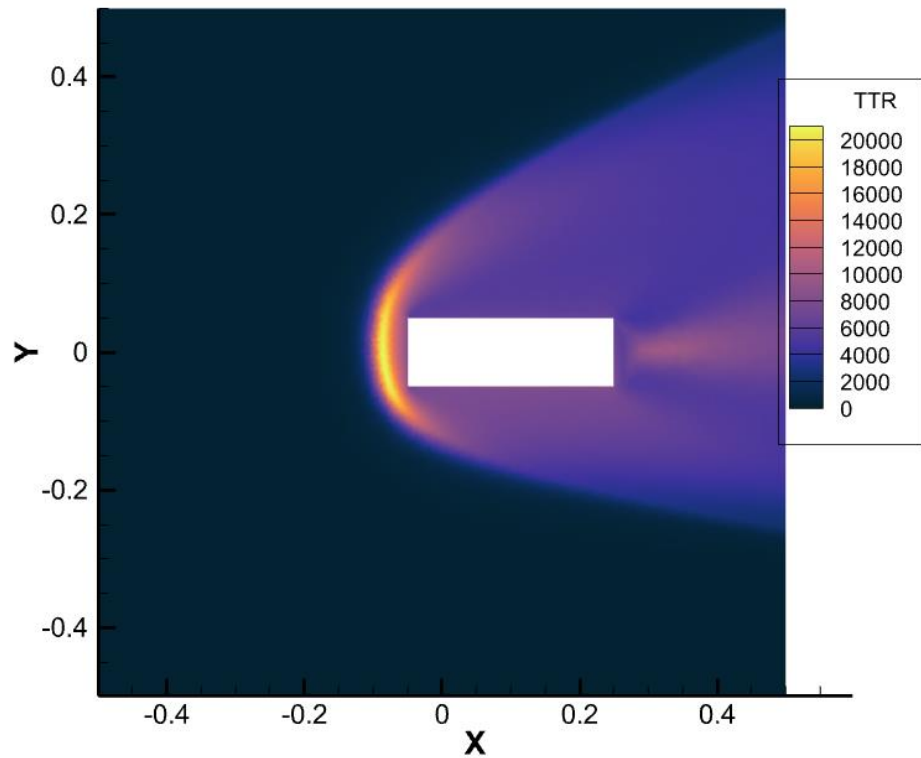


Figure 46: Translational Temperature Distribution of Mach 25 flow across 3U CubeSat, Side +Z face-pointing at 7.5° angle of attack

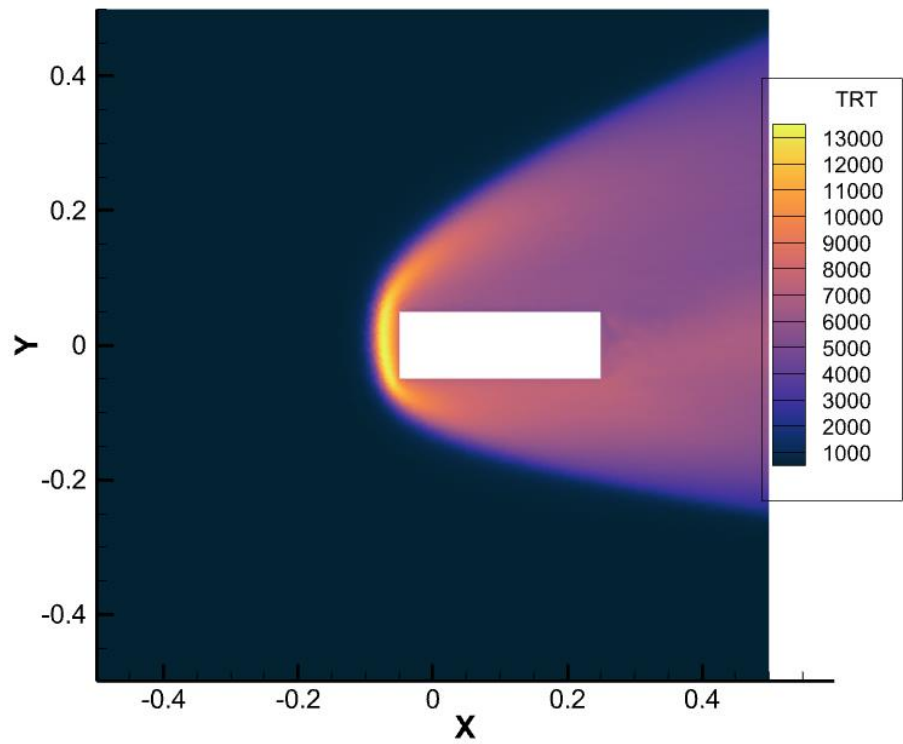


Figure 47: Rotational Temperature Distribution of Mach 25 flow across 3U CubeSat, Side +Z face-pointing at 7.5° angle of attack

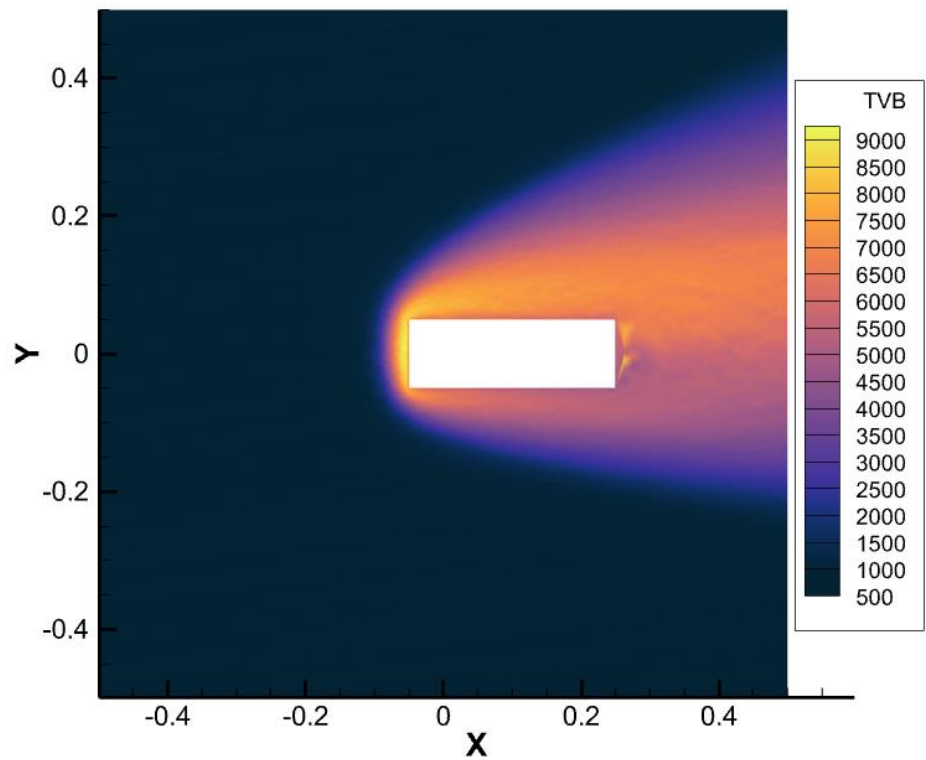


Figure 48: Vibrational Temperature Distribution of Mach 25 flow across 3U CubeSat, Side +Z face-pointing at 7.5° angle of attack

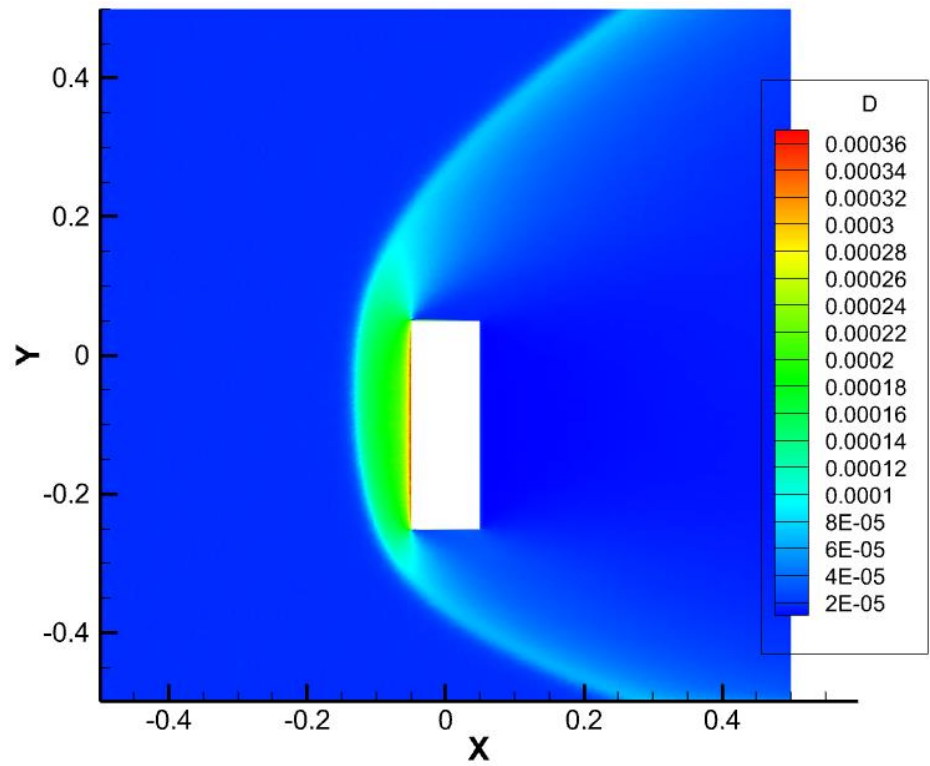


Figure 49: Density Distribution of Mach 25 flow across 3U CubeSat, Side +X or +Y face-pointing at 7.5° angle of attack

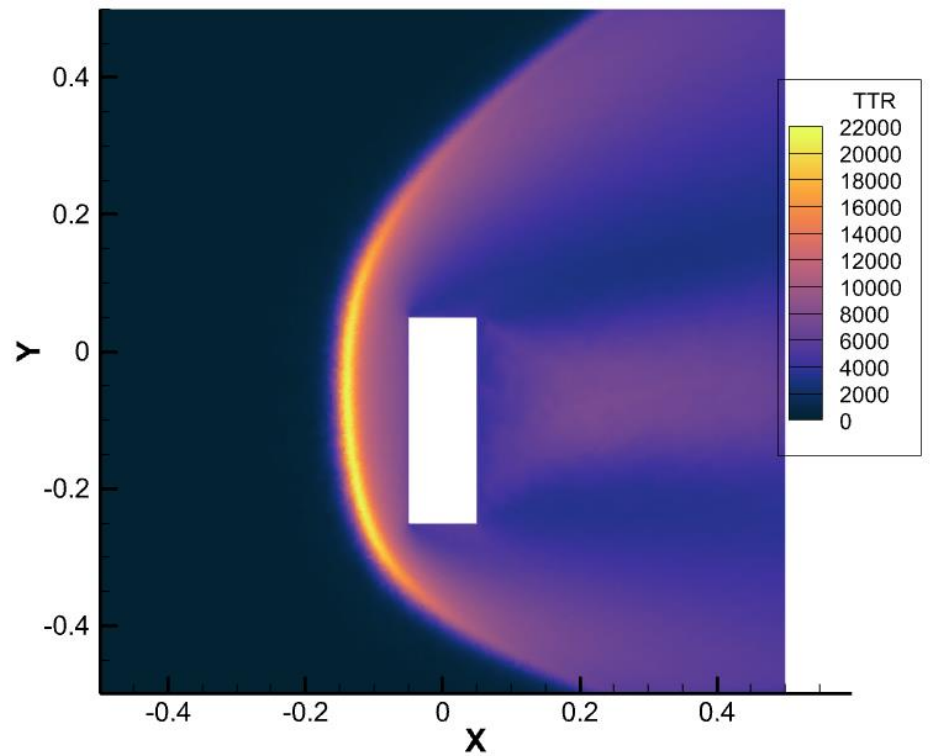


Figure 50: Translational Temperature Distribution of Mach 25 flow across 3U CubeSat, Side +X or +Y face-pointing at 7.5° angle of attack

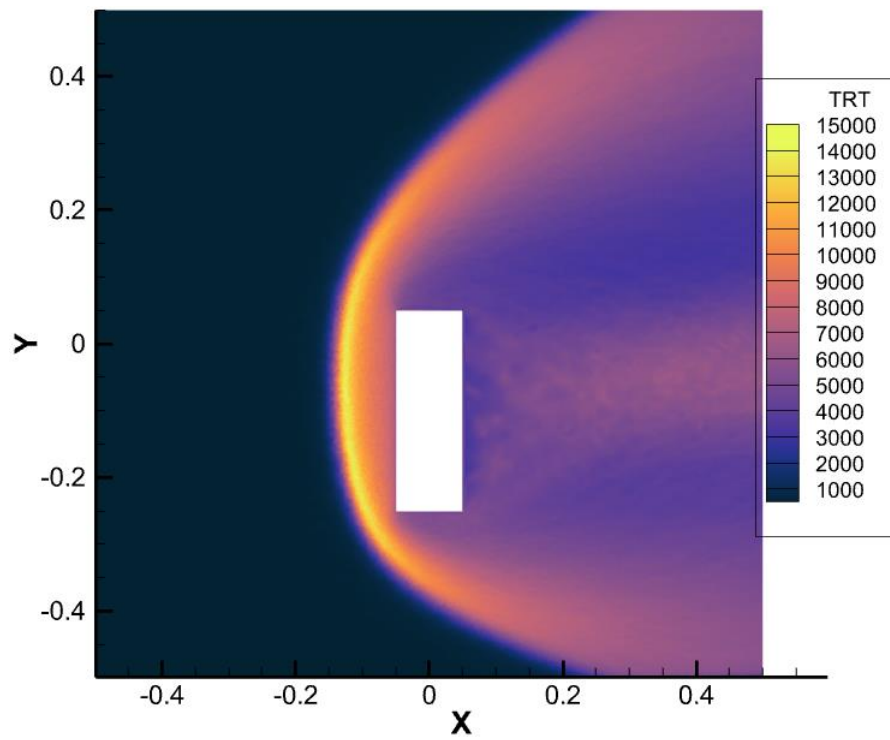


Figure 51: Rotational Temperature Distribution of Mach 25 flow across 3U CubeSat, Side +X or +Y face-pointing at 7.5° angle of attack

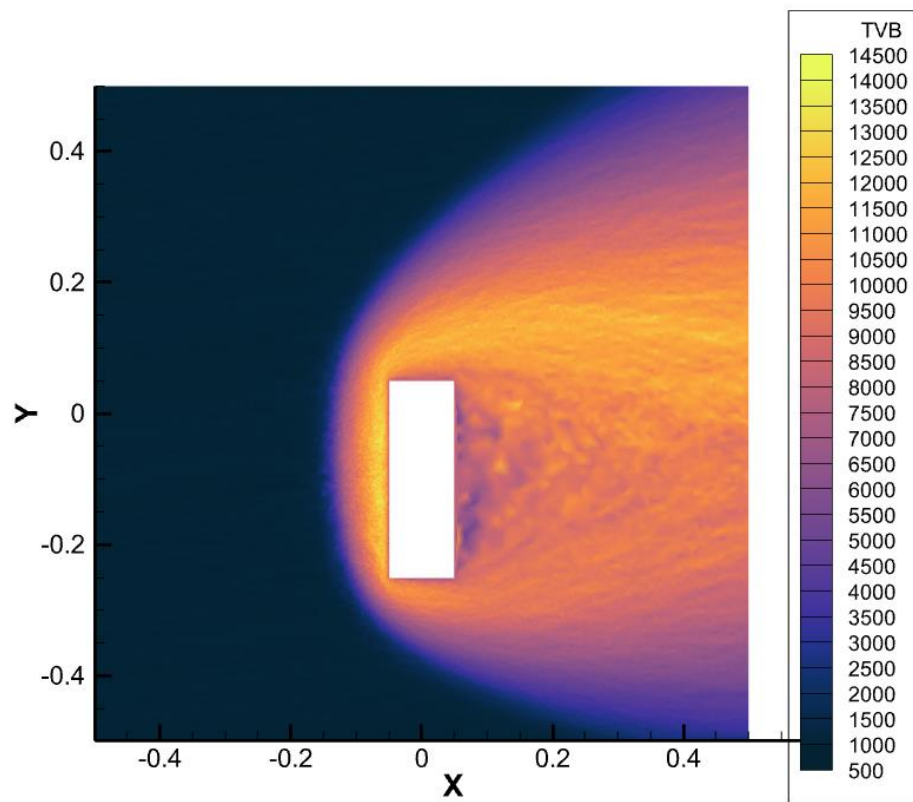


Figure 52: Vibrational Temperature Distribution of Mach 25 flow across 3U CubeSat, Side +X or +Y face-pointing at 7.5° angle of attack

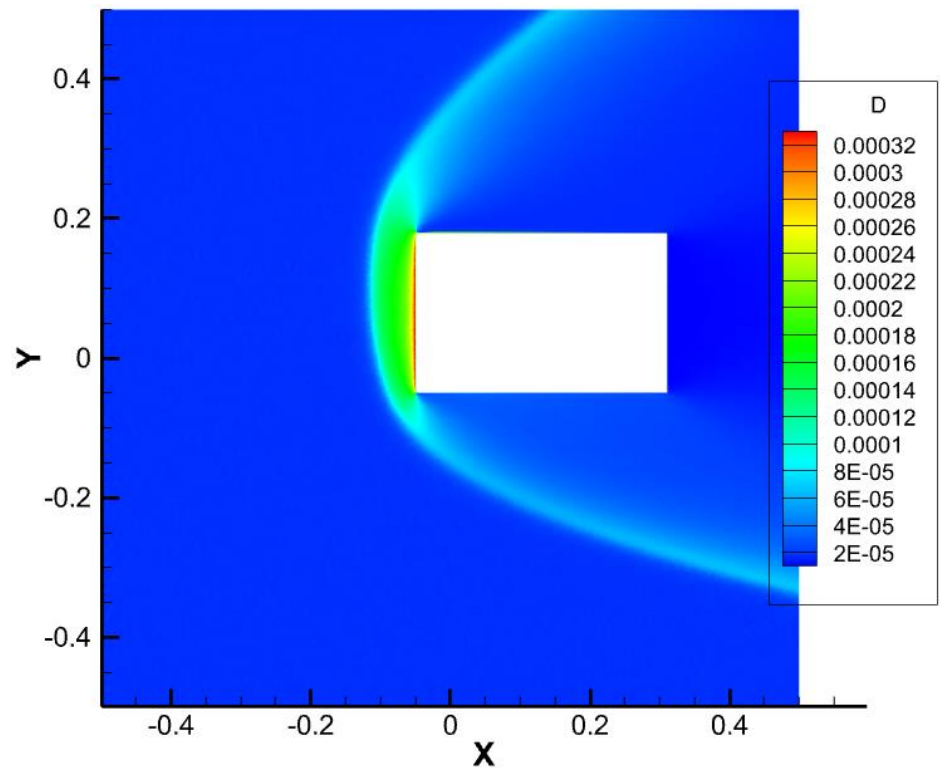


Figure 53: Density Distribution of Mach 25 flow across 6U CubeSat, Side +Z face-pointing at 7.5° angle of attack

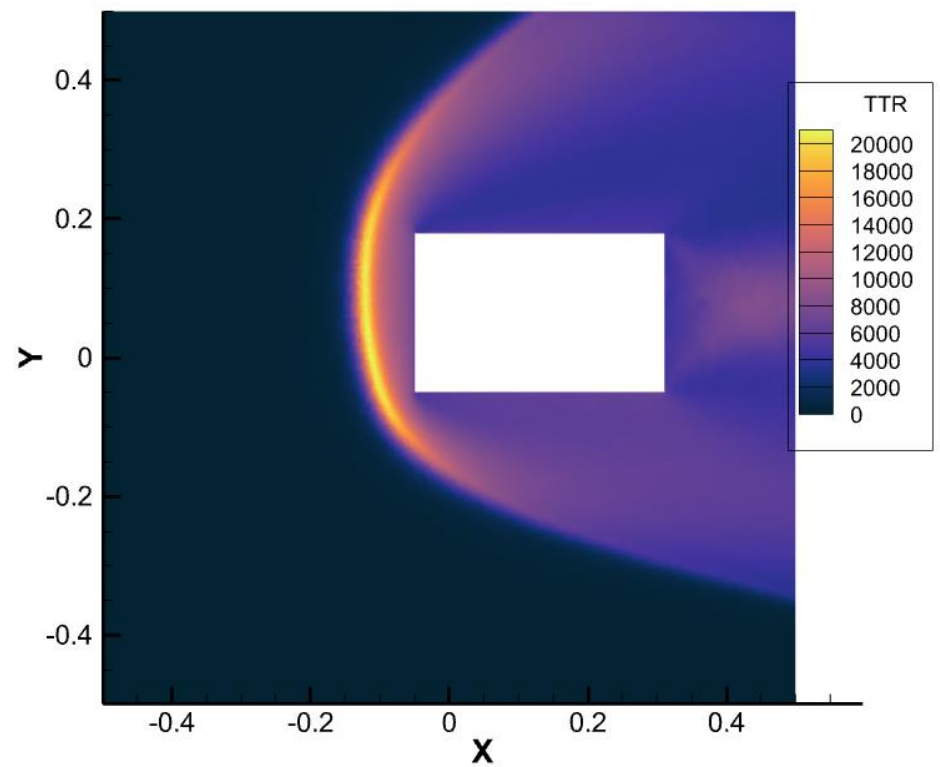


Figure 54: Translational Temperature Distribution of Mach 25 flow across 6U CubeSat, Side +Z face-pointing at 7.5° angle of attack

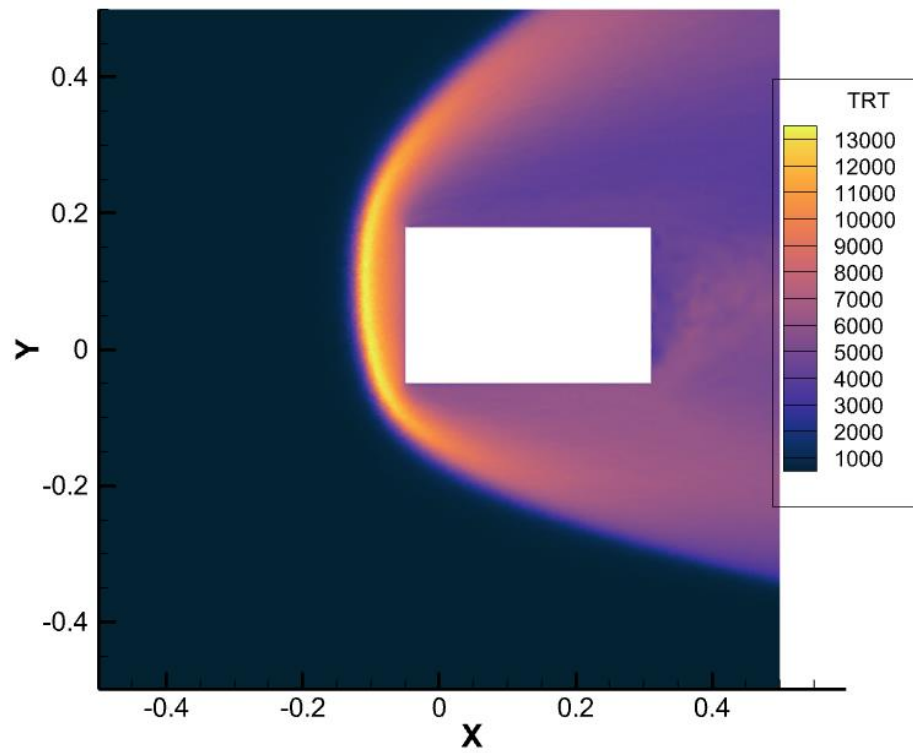


Figure 55: Rotational Temperature Distribution of Mach 25 flow across 6U CubeSat, Side +Z face-pointing at 7.5° angle of attack

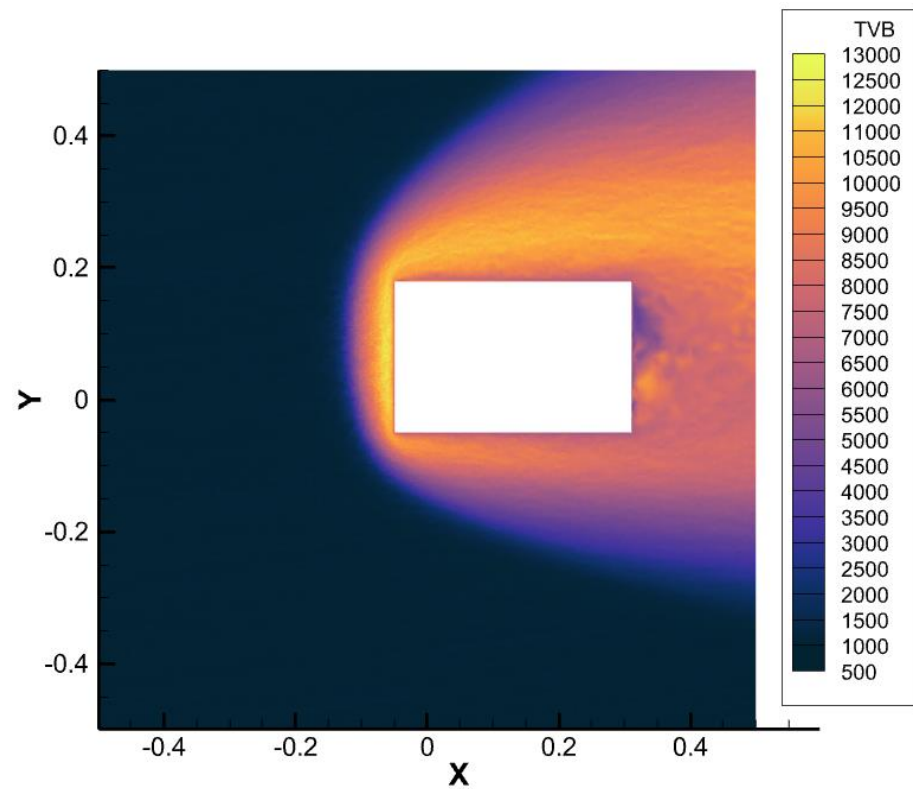


Figure 56: Vibrational Temperature Distribution of Mach 25 flow across 6U CubeSat, Side +Z face-pointing at 7.5° angle of attack

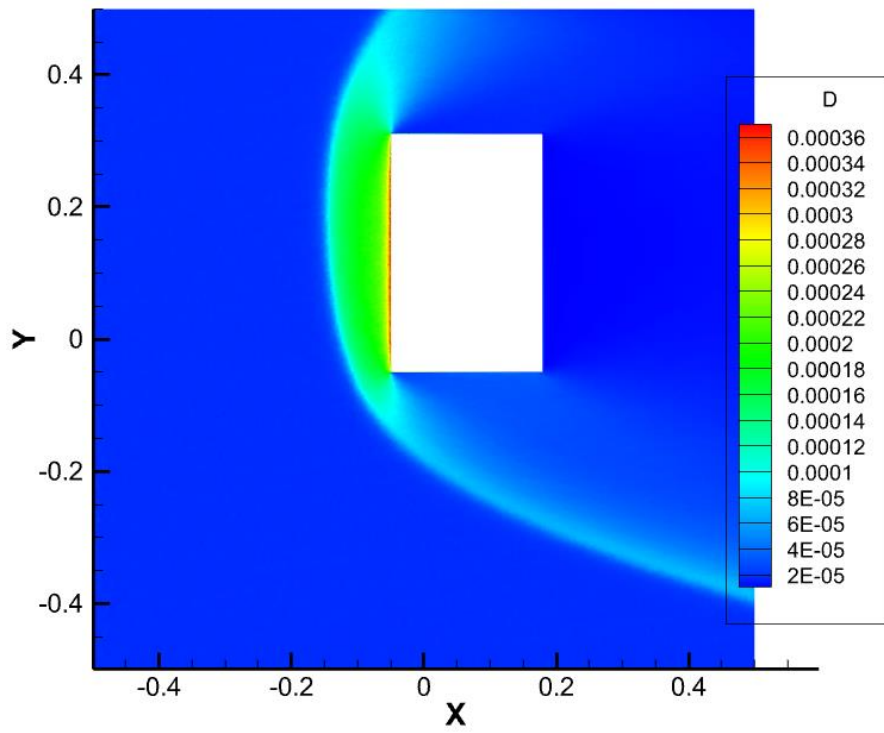


Figure 57: Translational Temperature Distribution of Mach 25 flow across 6U CubeSat, Side +X face-pointing at 7.5° angle of attack

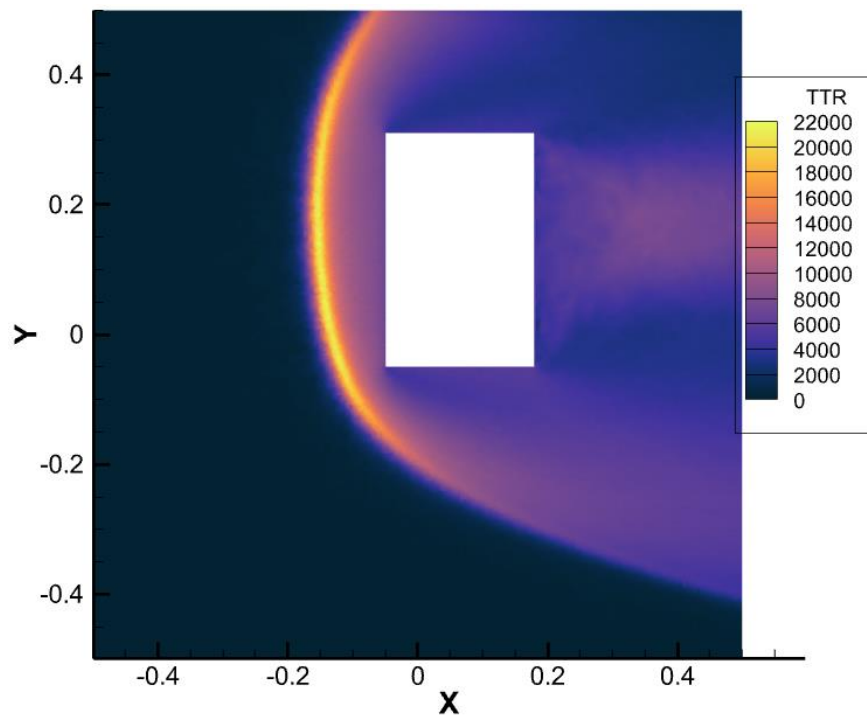


Figure 58: Translational Temperature Distribution of Mach 25 flow across 6U CubeSat, Side +X face-pointing at 7.5° angle of attack

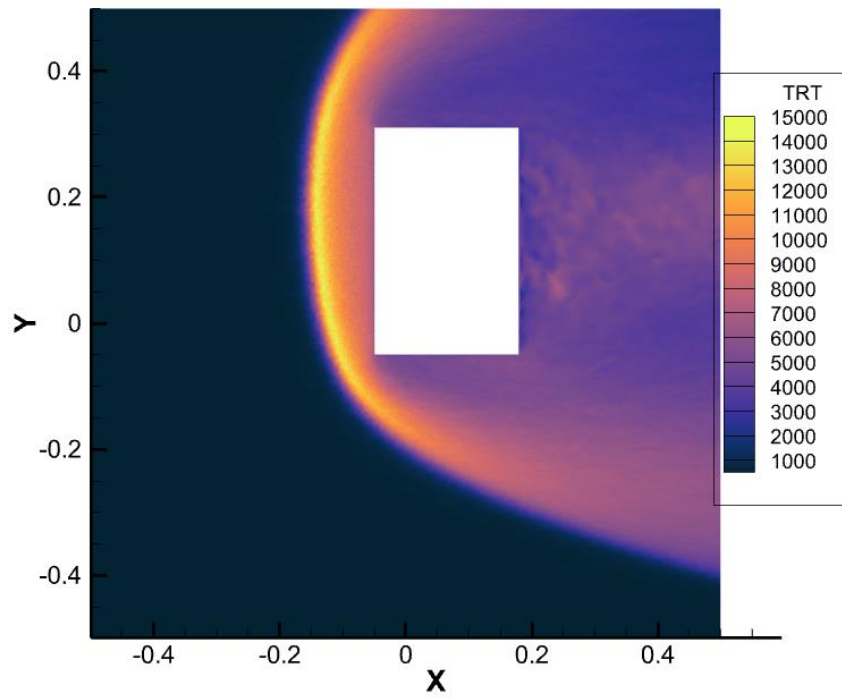


Figure 59: Translational Temperature Distribution of Mach 25 flow across 6U CubeSat, Side +X face-pointing at 7.5° angle of attack

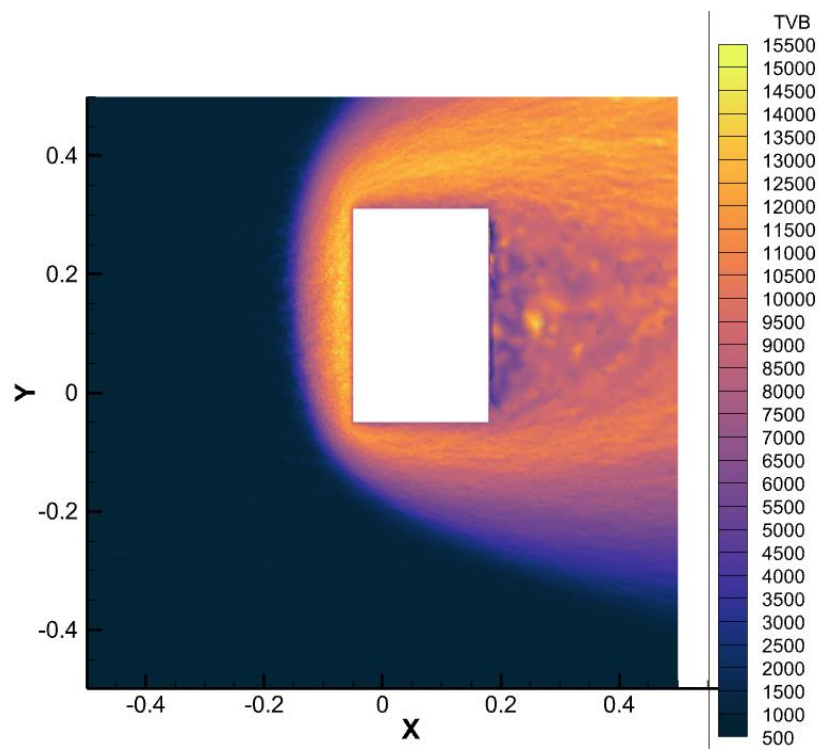


Figure 60: Vibrational Temperature Distribution of Mach 25 flow across 6U CubeSat, Side +X face-pointing at 7.5° angle of attack

3.4. Atmospheric Gas Models

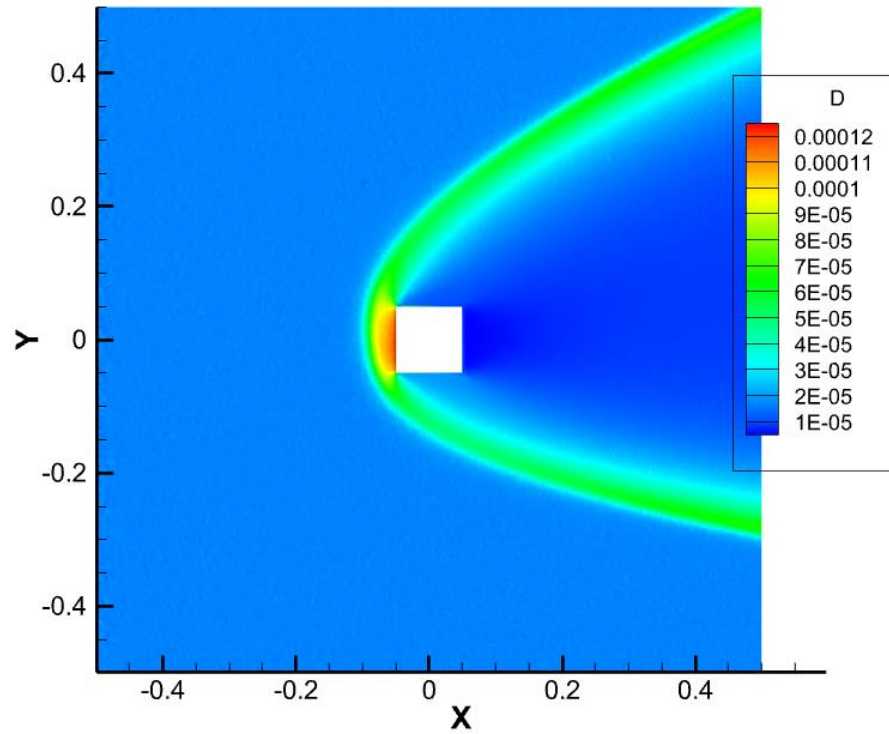


Figure 61: Density Distribution of Mach 25 flow across 1U CubeSat using ideal air model.

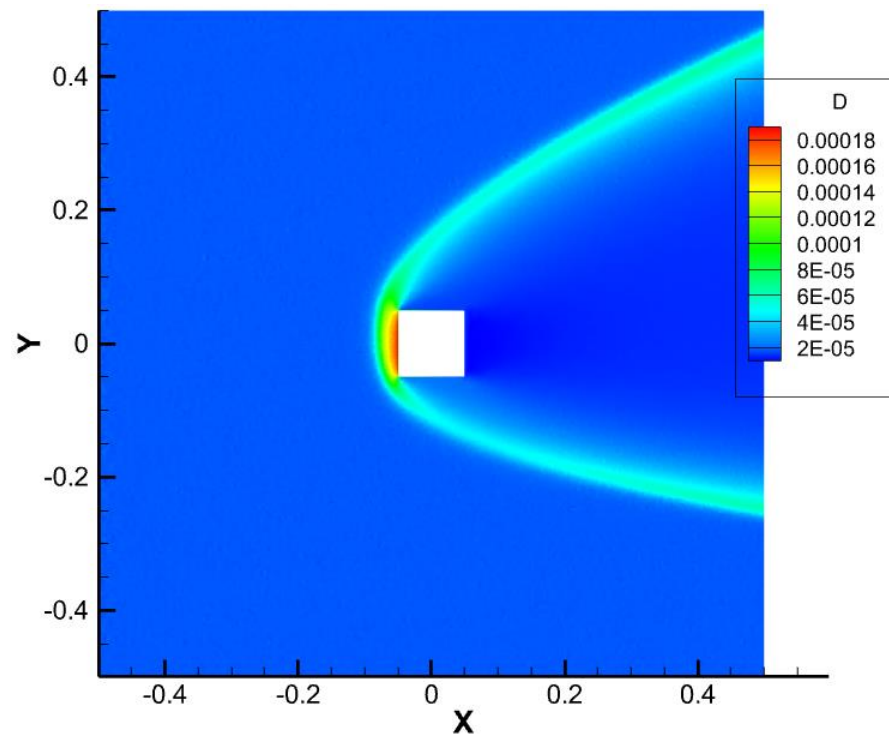


Figure 62: Density Distribution of Mach 25 flow across 1U CubeSat using real air 11-species model.

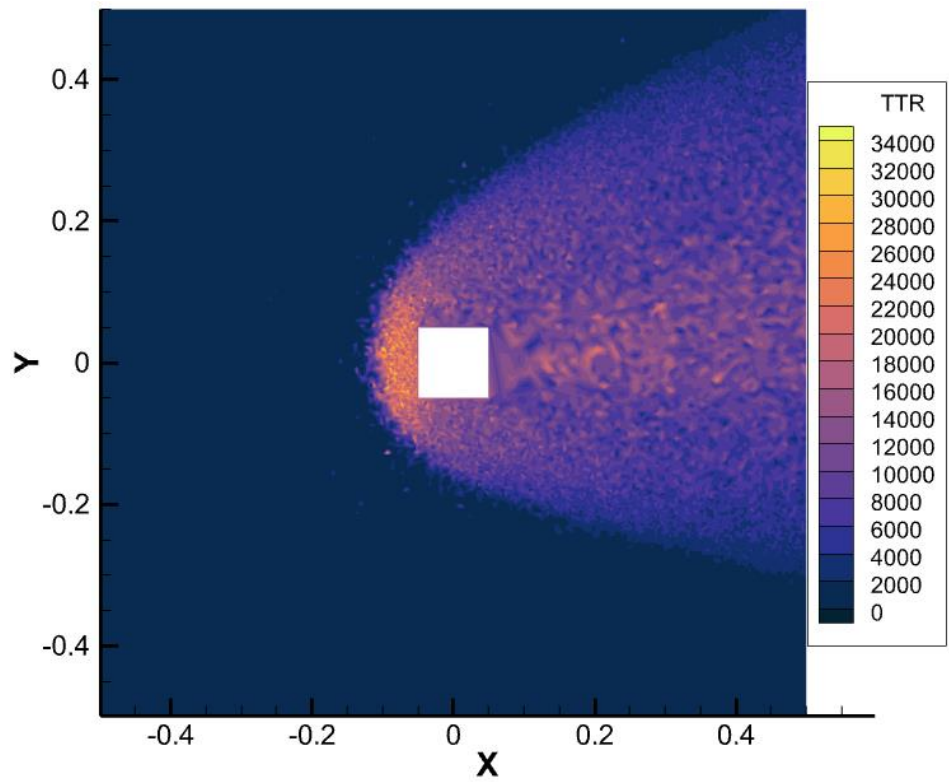


Figure 63: Translational Temperature Distribution of molecular oxygen species for Mach 25 flow across 1U CubeSat in ideal air.

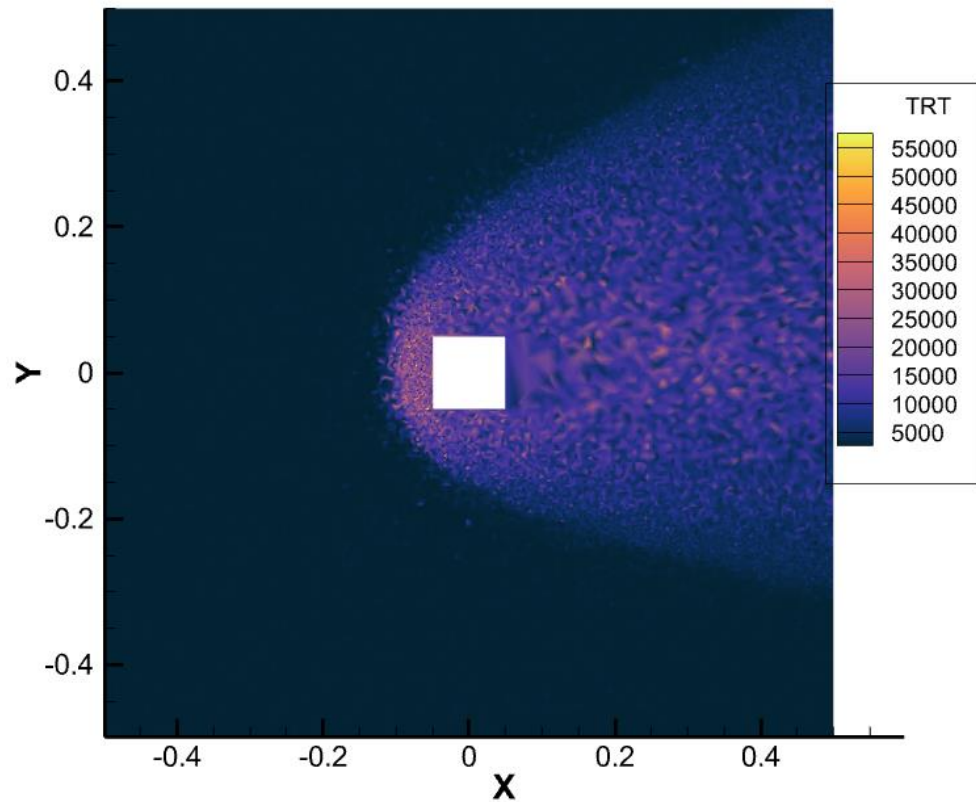


Figure 64: Rotational Temperature Distribution of molecular oxygen species for Mach 25 flow across 1U CubeSat in ideal air.

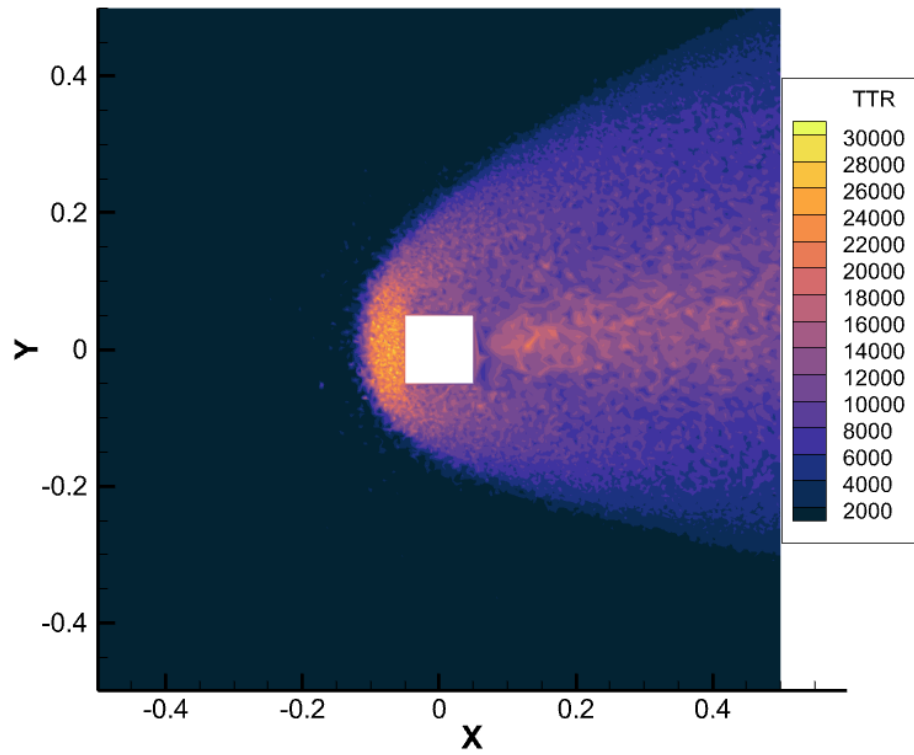


Figure 65: Translational Temperature Distribution of molecular nitrogen species for Mach 25 flow across 1U CubeSat in ideal air.

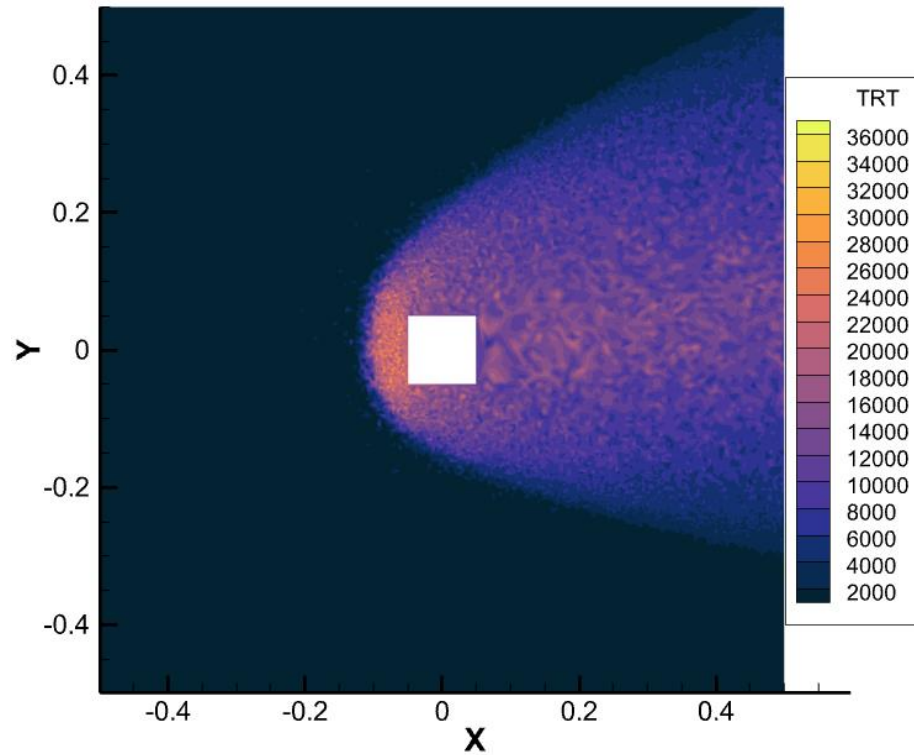


Figure 66: Rotational Temperature Distribution of molecular nitrogen species for Mach 25 flow across 1U CubeSat in ideal air.

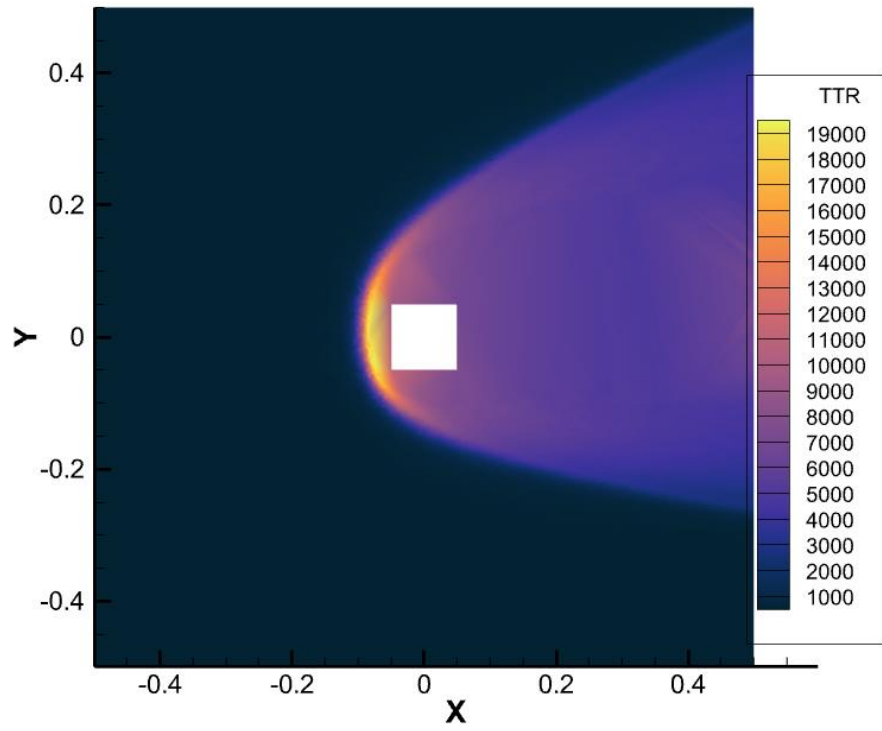


Figure 67: Translational Temperature Distribution of molecular oxygen species for Mach 25 flow across 1U CubeSat in real air.

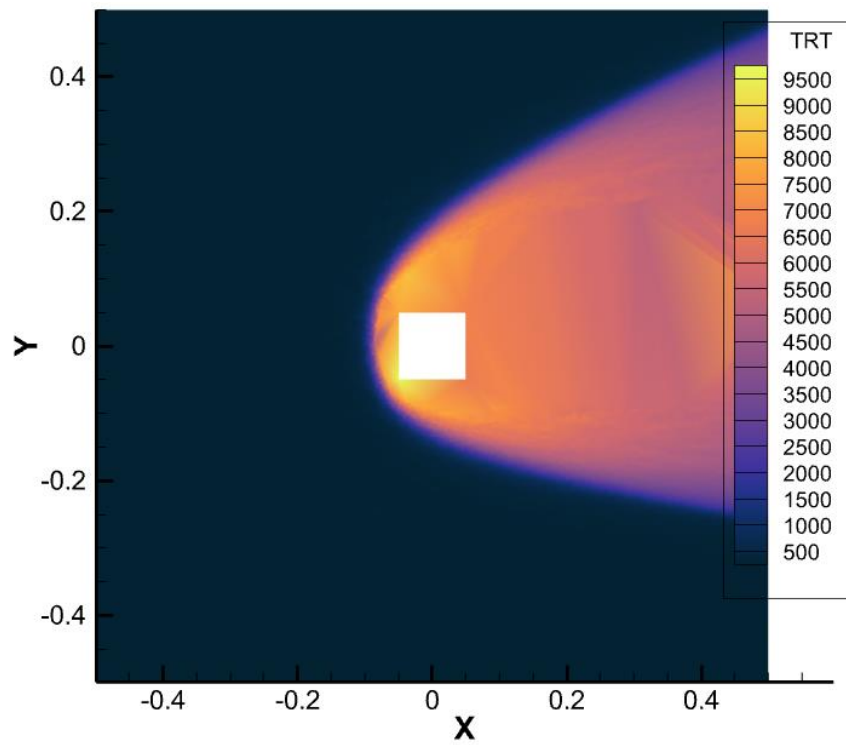


Figure 68: Rotational Temperature Distribution of molecular oxygen species for Mach 25 flow across 1U CubeSat in real air.

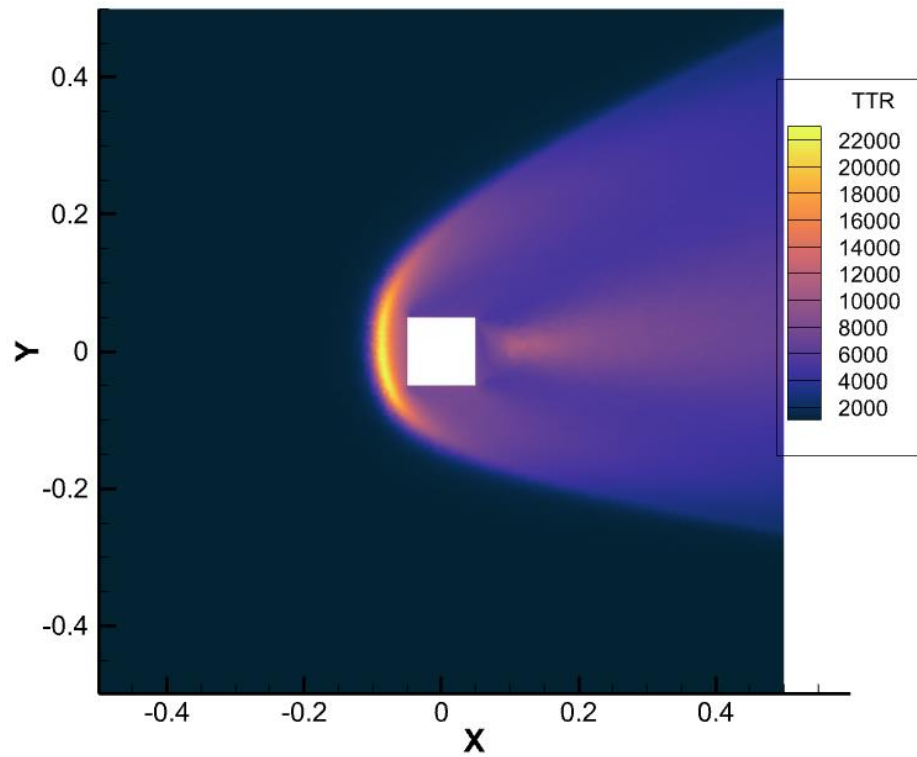


Figure 69: Translational Temperature Distribution of molecular nitrogen species for Mach 25 flow across 1U CubeSat in real air.

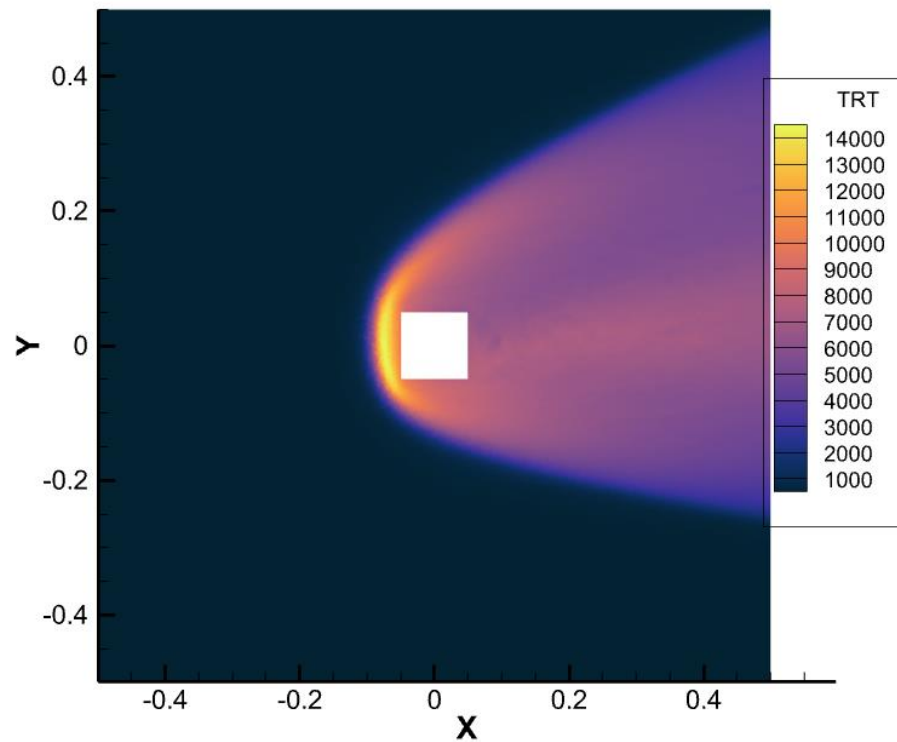


Figure 70: Rotational Temperature Distribution of molecular nitrogen species for Mach 25 flow across 1U CubeSat in real air.

4. Discussion

4.1. General Discussion

As previously discussed in Section 1.2.2.1. Rarefied Gas Dynamics, the hypersonic flow is in the rarefied regime at high Knudsen numbers, around 0.1 to 100. Due to the breakdown of the continuum assumption, CFD calculation is not a suitable method for drag analysis. This is further supported by a study done by Holman and Boyd in 2011 [56].

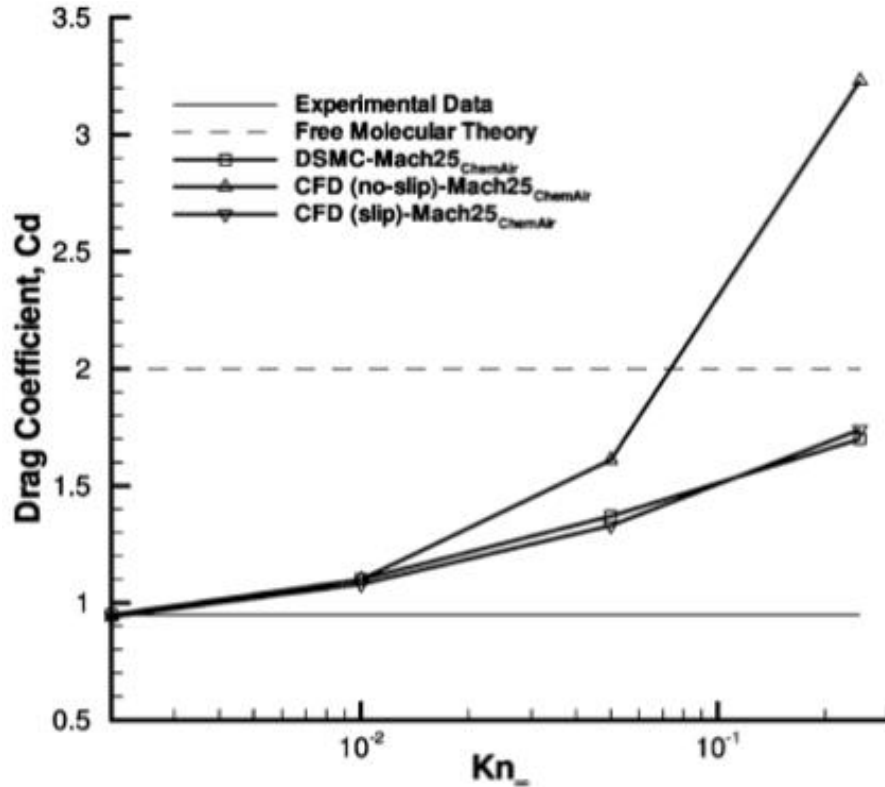


Figure 71: Graph of Drag Coefficient C_d against Knudsen number of the freestream Kn_∞ comparing CFD and DSMC results for hypersonic real air flow across a sphere at Mach 25 [56].

From Figure 71, it is evident that the CFD solution with no-slip boundary conditions overpredicts the value of the drag coefficient at Knudsen numbers greater than 0.05. This is consistent with the literature review in Section 1.2.2.1. Rarefied Gas Dynamics, and the Knudsen number values obtained from the breakdown parameter by Bird [19]. The CFD solution with slip boundary conditions, defined by velocity slip and temperature jump, is closer to the solution given by DSMC. Similarly, the CFD solution have overpredicted the vibrational and rotational temperature distributions caused due to thermal nonequilibrium, when compared with the DSMC solution [57]. Holman and Boyd have also stated that we will obtain an unrealistic value for heat flux from CFD solutions when compared with DSMC, leading to earlier separation of boundary layer and decreased stream temperature downstream [56].

We ran simulations to study the effect of angle of attack on the drag coefficient by altering the x and y components of the velocity. Alternatively, this study can be conducted by redefining the geometrical parameters for a given angle of attack. The resultant drag coefficients have been illustrated as follows:

Angle of Attack (°)	Drag Coefficient C_D	Error %
0.00	0.1879	5.101%
2.50	0.1883	4.899%
5.00	0.1847	6.717%
7.50	0.1880	5.051%
10.00	0.1858	6.162%

Table 9: Drag Coefficients for different angle of attack for a 1U CubeSat.

The values of the drag coefficients vary from the analytical result of 0.198 within a range of error percentage between 5% to 7%. The range of error percentage described for the drag coefficients is a widely accepted range, thus validating our DSMC solutions for the simulations under this category[19].

4.2.2. Solar Activity

Depending on the level of the solar activity, the short wavelength solar electromagnetic (EM) radiation can vary significantly. The short wavelength solar EM radiation consists of waves from the extreme ultraviolet (EUV) and ultraviolet (UV) spectra. The solar activity can change the density of the air present in the lower thermosphere. During increased solar activity, referred to as Extremely High Solar Activity, the enhanced interaction between Earth's magnetic field and solar wind can cause thermochemical fluctuations [59]. The MSISE-90 model of the Earth's upper atmosphere is based on data obtained from a variety of mass spectrometers on board different satellites and incoherent scatter radar data [60, 61].

Using the MSISE-90 model, the data for temperature, density and pressure for a given level of solar activity have been provided in Table 10.

Level of Solar Activity	Temperature (K)	Pressure (Pa)	Density (kg/m^3)
Low	203.1065	0.981	1.69×10^{-5}
Mean	196.3636	0.945	1.68×10^{-5}
Extremely High	172.2146	0.842	1.68×10^{-5}

Table 10: Temperature, Pressure and Density of Air at altitudes from 80 – 100 km for variable solar activity [62].

There were three simulations that were modelled to obtain the drag coefficients for varying level of solar activity, based on the MSISE-90 model. The resultant drag coefficients have been provided in Table 11.

Level of Solar Activity	Drag Coefficient C_D	Error%
Low	0.1872	5.454%
Mean	0.187	5.556%
Extremely High	0.1856	6.262%

Table 11: Drag coefficients for different form of solar activity.

Despite the small differences between the drag coefficients, it is evident that change in the pressure, temperature and density of the free stream atmospheric air can affect the drag coefficients. The most notable change can be noticed where the density underwent a change in the order of 10^{-7} kg/m³ under extremely high solar activity, resulting in a jump in the error percentage by approximately 1%. The increase in the error percentage is significant considering the density change is quite small. The change can be further validated by the clear difference in the values of the drag coefficient under low and extremely high solar activity.

Since the drag force acting on the CubeSat is dependent on the density of the reference free stream gas (atmospheric air, in this case), it is obvious to say that the solar activity affects the drag coefficient by actively changing the reference equilibrium state. The drag of the CubeSat increases at higher altitudes in the atmosphere because of an increase in the intensity of the effects of solar activity [63].

4.2.3. Geometry of CubeSat

The Geometry of CubeSat. The dimensions of the geometry of the CubeSats have been described by

CubeSat Configuration	Dimensions (cm)
1U	10 × 10 × 10
2U	20 × 10 × 10
3U	30 × 10 × 10
6U	34 × 20 × 10

Table 12: Dimensions for different CubeSat geometries [64].

The different CubeSat geometries have been illustrated here

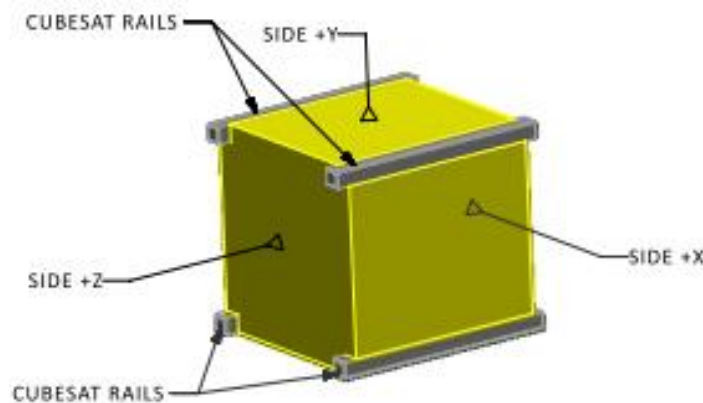


Figure 74: 1U CubeSat Geometry, describing +X, +Y and +Z side of the satellite[64].

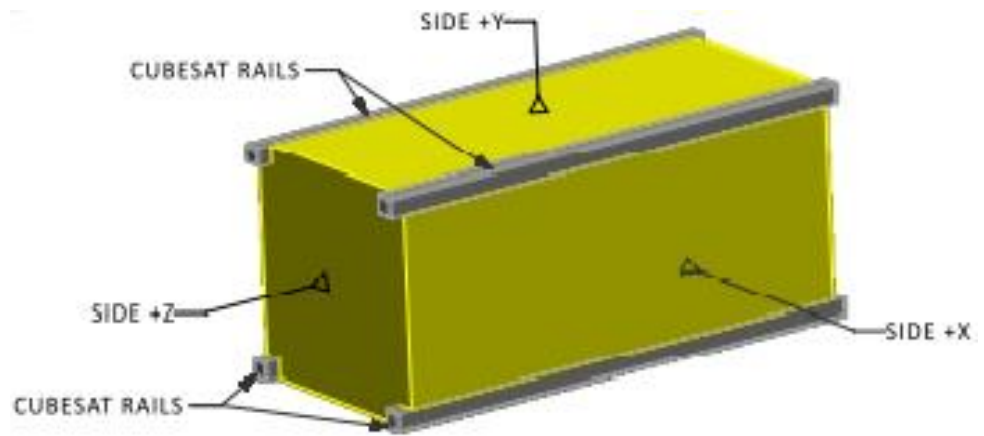


Figure 75: 2U CubeSat Geometry, describing +X, +Y and +Z side of the satellite[64].

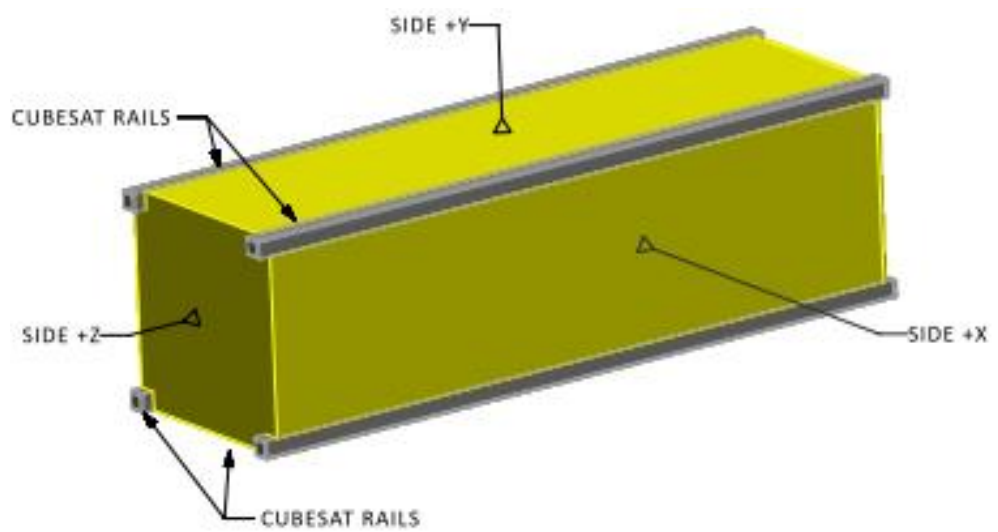


Figure 76: 3U CubeSat Geometry, describing +X, +Y and +Z side of the satellite[64].

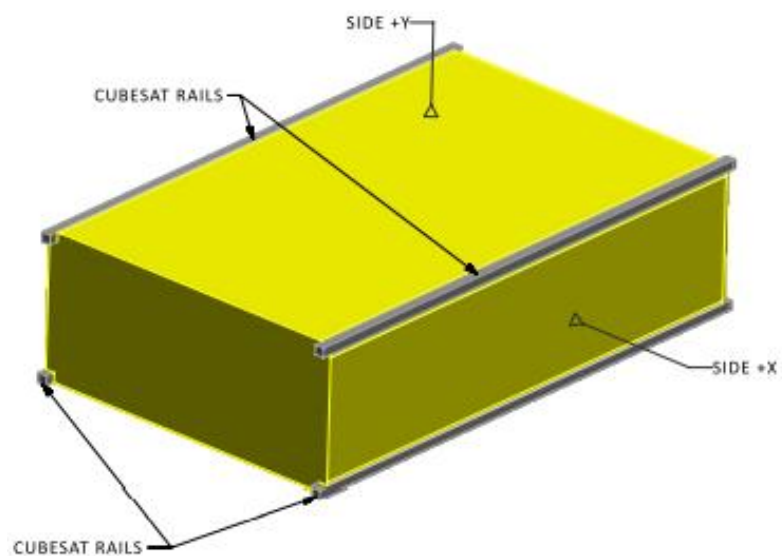


Figure 77: 6U CubeSat Geometry, describing +X, +Y and +Z side of the satellite[64].

We have studied the effect of the geometry of the CubeSat and its orientation with the atmospheric air by modelling simulations using the input parameters stated in Sections 2.3. **Input Parameters for DSMC Calculations in**

DS2V and 2.4. **Surface Modelling and Geometry Parameters** The resulting drag coefficients have been compared with their respective analytical prediction for C_D from Section 1.2.3. Aerodynamic Drag in Rarefied Gas to obtain the error percentage for validation of results and analysis.

CubeSat Configuration	Surface Dimensions (cm)	Drag Coefficient C_D	Analytical Prediction for C_D	Error%
1U	10 × 10	0.1880	0.1980	5.051%
2U	10 × 20	0.1926	0.1980	2.727%
2U	20 × 10	0.3756	0.3960	5.151%
3U	10 × 30	0.1970	0.1980	0.505%
3U	30 × 10	0.5758	0.5940	3.064%
6U	22 × 34	0.4340	0.4356	0.367%
6U	34 × 22	0.6614	0.6732	1.753%

Table 13: Drag coefficients for different orientations of CubeSat geometries.

It is evident that initially the analytical prediction for C_D is an overprediction in comparison to the drag coefficients obtained from the DSMC calculations. Also, it is easy to overlook the error percentage as they are well within the limit. However, from the error percentage, we can observe that as we increase the surface dimensions, the drag coefficient predictions by the DSMC calculations start getting closer to the theoretical / analytical prediction. This reinforces our belief that the DSMC method is more appropriate method as compared to CFD, as discussed in Section 4.1. **General Discussion.**

4.2.4. Thermochemical Nonequilibrium

We performed simulations to obtain drag coefficients for the different gas models / chemical kinetic models to study the effect of chemical nonequilibrium on drag coefficient. The drag coefficients of the three simulations have been tabulated below alongside their respective chemical kinetic models.

Air Model	Drag Coefficient C_D	Error%
Ideal Air	0.1802 (equilibrium)	–
Real Air, 5-species	0.1880 (nonequilibrium)	5.051%
Real Air, 11-species	0.1902 (nonequilibrium)	3.940%

Table 14: Drag Coefficient for different models for atmospheric air for 1U CubeSat.

The calculated drag coefficient C_D is smaller when the atmospheric air is in equilibrium, as compared to the ones where the air is in a state of nonequilibrium. The ideal air model does not have any vibrational modes for its two molecular species – molecular nitrogen and molecular oxygen. The lack of vibrational modes

renders the ideal air model incapable of modelling vibrational nonequilibrium. It becomes evident that the vibrational temperature affects the drag coefficient. Similarly, the ideal air model lacks dissociation reactions in the viscous boundary layer ahead of the body. The additional molecular species in the real air models introduce fluctuations in the molecular weight. As a result, drag coefficient of the CubeSat can be affected by chemical nonequilibrium too. The low error percentages for the C_D obtained using the real air 5-species and 11-species model confirm that the 5-species and the 11-species model are ideal for atmospheric re-entry analysis, as the model considers the effects of thermochemical nonequilibrium.

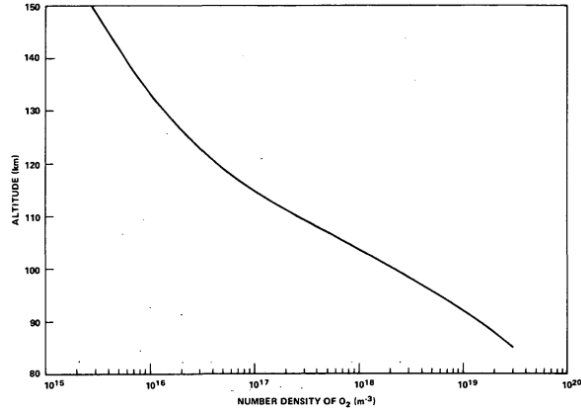


Figure 78: Graph depicting the variation of number density of O_2 between altitudes 86 km to 150 km [62].

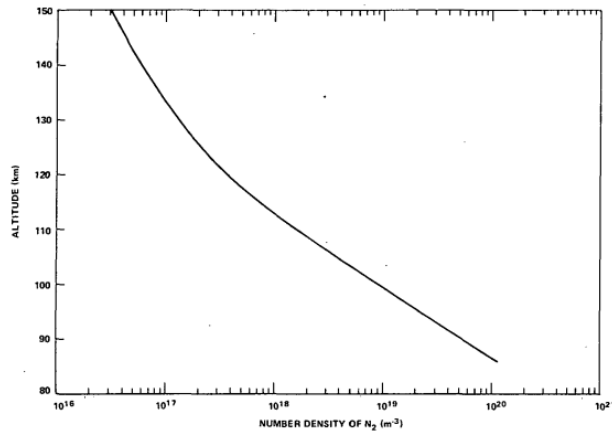


Figure 79: Graph depicting the variation of number density of N_2 between altitudes 86 km to 150 km [62].

From Figure 61 to Figure 70, it is evident that the ideal air model underpredicts the density distribution across the surface of the CubeSat, while giving unrealistic predictions of translational and rotational nonequilibrium for molecular oxygen and molecular nitrogen species. This is due to lack of the presence of atomic species and nitric oxide in the ideal air. The unrealistic predictions are further aided by the inability of modelling vibrational nonequilibrium using the ideal air model. This further puts emphasis on the fact that the real air 5-species or 11-species model is best suited for hypersonic re-entry studies. The number density distributions for

molecular oxygen and molecular nitrogen were found to be consistent with the results in Figure 78 and *Error! Reference source not found.*, and comparative studies done by Boyd and Gokcen [65], and Kim et al [18].

4.3. Limitations

As previously described in Section 2.1.3. DS2V Solver, DSMC results do not converge because of the simulated hypersonic flow is unsteady in nature with respect to real physical time. Hence, to mitigate the risk of using poor datasets for analysis, a Python code was written to test for “convergence” of the DSMC solution, as shown in Appendix A. The “convergence” of the DSMC solution was compared with the standard deviation in the results from the original output from DS2V.

Computational approximations can range from the number of simulated molecules in each computational cell, collision sampling procedure, cell size and time step. The drag coefficient can be plotted as a function of number of simulated molecules in a computational cell. Acceptable results are obtained for an average of at least ten molecules per cell in the most critical region of the flow. Major errors are obtained when the number of molecules per cell goes below five [19, 53].

There are three basic procedures that the DS2 program uses for collision sampling in DSMC calculations:

1. Typical collision pairs are selected according to appropriate probabilities associated with each cell.
2. The non-equilibrium collision rate in each cell is computed from all probable collision pairs.
3. All collision pairs are considered, and a collision is computed with probability equal to the ratio of the volume swept out by the collision cross-section

Because of the necessity to check all potential collision pairs, procedures 2 and 3 have a computational time which is proportional to the square of average number of molecules per cell. A typical dimension of the cells in the standard calculation is nearly equal to the local mean free path. It has been observed that the drag coefficient C_D increases if an unrealistic dimension for cell size is used. One of the most critical dimensions is the height of the cell in the “boundary layer” region adjacent to the edges of the CubeSat. By adjusting the cell size near the boundary layer, one can achieve an accurate result for C_D [43].

The usual criterion for the time step Δt is that a molecule should move about a quarter of the cell dimension during one step. The average time step for the study was approximately 5×10^{-7} second. When the distance moved by the simulated molecule during a time step becomes larger than the cell size, we obtain erroneous data as a significant increase in C_D is observed. However, a small reduction in C_D is noticed if the time step is less than the normal value by order of magnitude. Nevertheless, the small reduction observed is negligible [21].

5. Conclusions and Further Work

5.1 Conclusions

This study investigated how aerodynamic properties like the drag coefficient is affected by the breakdown of continuum assumptions in rarefied gas flow. As the results are consistent with the literature cited in this work, we can conclude that thermochemical nonequilibrium, solar activity, the geometry of the CubeSat and the angle of attack are driving parameters for future CubeSat re-entry analysis, as these factors have a direct influence on the drag coefficient of the spacecraft.

We can also conclude that the Direct Simulation Monte Carlo (DSMC) makes accurate predictions for the drag coefficients of a spacecraft in rarefied gas regime, when compared with the CFD method. We can also draw this conclusion that the real air 5-species and 11-species model give an accurate prediction of the thermal and chemical nonequilibrium, when compared to the ideal air model. Overprediction of the drag coefficient by CFD and overprediction of the thermochemical nonequilibrium by the ideal air model can have significant effect on the design of CubeSat, which can affect space mission analysis and design (SMAD) for a CubeSat, leading to a compromise in the mission requirements and objectives.

5.2 Further Work

The results presented in this study will be extended to three-dimensional flow and compared with the DSMC calculations obtained from other DSMC solvers like SPARTA and dsmcFOAM. This will also allow us to compare the computational performance of DSMC calculations performed using serial computing (e.g., DS2V) and those performed using parallel computing (e.g., SPARTA and dsmcFOAM). The results can also be compared with re-entry analysis tools like NASA Debris Assessment Software (DAS) and Object Re-entry Survival Analysis Tool (ORSAT), and ESA Spacecraft Atmospheric Re-entry and Aerothermal Breakup (SCARAB) and Spacecraft Entry Survival Analysis Module (SESAM). By comparing the results in this study against the results of the re-entry analysis tools, we can make recommendations on how we can improve the survivability of the CubeSats to enable interplanetary sample return missions. The study allows the scope for extending the drag analysis to a dynamic aerothermodynamic study where the CubeSat is in tumbling motion during re-entry. The results can be compared with experimental data for further validation of the analysis presented, thus enabling us to design a unique low-cost solution microelectromechanical system (MEMS) based attitude control subsystem (ACS) for CubeSats doing missions on low Earth orbit (LEO).

Appendix A

Convergence.py Python code used to check “convergence” of DSMC solution

```
import time

import numpy as np

converg=0

density_ref=[0,0]

while converg == 0:

    file = open("DS2FF.DAT", 'r')

    lines = file.readlines()

    file.close()

    density=[]

    nlines=lines[30:3500]

    ref0=np.asarray(density_ref[0], dtype=np.float64)

    for line in nlines:

        parts = line.split() # split line into parts

        x = parts[3]

        density.append(x)

    #print(density)

    density_ref[1]=(density)

    #print(density_ref[1])

    if density==density_ref[0]:

        print('CONVERGENCE')

        converg=1

    else:

        density_ref[0]=density

        #print(density_ref[0][0:10])

        time.sleep(360)

    ref1=np.asarray(density_ref[1], dtype=np.float64)
```

```
dif=np.subtract(ref0,ref1)

#print(ref0,ref1)

#print(dif)

print(np.count_nonzero(dif))

# result = all(element == density[0] for element in density)

# if (result):

#   converg=1

#   print('CONVERGENCE')
```

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